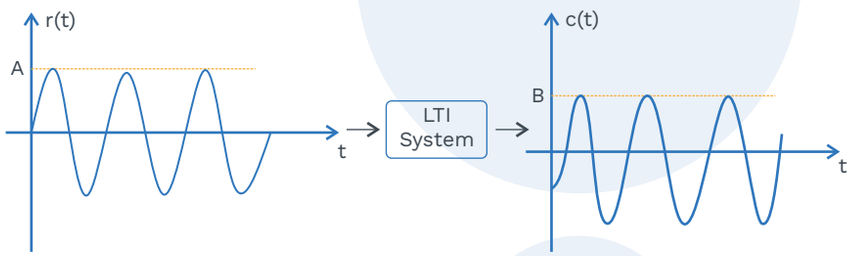
# 7 Frequency Response Analysis



## INTRODUCTION:

- The frequency response of a system is defined as the magnitude and phase relation between the sinusoidal input and the steady state output of a system.
- Frequency response is independent of the amplitude and phase of the input signal.
  In a linear time invariant system, if the input signal is –
  $r(t) = A \sin \omega t$
  Then, the steady state output can be written as –
  $c(t) = B \sin (\omega t + \phi)$
- The sinusoidal response differs from the input waveform only in magnitude and phase angle.




## CONCEPT OF FREQUENCY RESPONSE –

Let us consider a linear time invariant system having open loop transfer function G(s).

$$G(s) = \frac{C(s)}{R(s)} \quad \text{.....(I)}$$

If the input signal is –

$$r(t) = A \sin \omega t$$

So, the steady state response of a stable LTI system to a sinusoidal input can be written as –

$$C(s) = G(s)\, R(s)$$

Put $s = j\omega$,

$$C(j\omega) = G(j\omega)\, R(j\omega)$$

Thus, the output is the product of the input and $|G(s)|$.

**Rack your Brain**

What are the advantages of frequency response of a linear time invariant (LTI) system?

- Sinusoidal transfer function G(jω) can be obtained by substituting s = jω in equation (I), we get –

$$G(j\omega) = \frac{C(j\omega)}{R(j\omega)}$$

or $G(j\omega) = M\angle\phi$

Where, M = Gain or magnitude ratio

$\phi$ = Angle by which output leads the input

and $M = |G(j\omega)| = \frac{|C(j\omega)|}{|R(j\omega)|}$

$$\phi = \angle G(j\omega) = \angle C(j\omega) - \angle R(j\omega)$$

**Gray Matter Alert!!!**

For stable system, frequency of sinusoidal output must be same as the input frequency, but magnitude and phase of the output will be different from the input.

**Example – 1**

Calculate the steady state output, if the input is x(t) = sin2t and the transfer function of the system is

$$H(s) = \frac{1}{s+5}$$

**Solution:**

Given transfer function is –

$$H(s) = \frac{1}{s+5}$$

Put s = jω,

$$H(j\omega) = \frac{1}{5+j\omega}$$

$$x(t) = \sin 2t$$

$$\omega_0 = 2$$

$$H(j2) = \frac{1}{5+j2} = \frac{1}{\sqrt{29}}\angle -21.8^\circ$$

Output of the system is –

$$y(t) = |H(j\omega_0)| \sin(2t + \angle H(j\omega_0))$$

$$y(t) = \frac{1}{\sqrt{29}} \sin(2t - 21.8^\circ)$$

**Previous Year's Questions**

A system with transfer function $\frac{Y(s)}{X(s)} = \frac{s}{s+p}$ has an output $y(t) = \cos\left(2t - \frac{\pi}{3}\right)$ for the input signal $x(t) = p\cos\left(2t - \frac{\pi}{2}\right)$, then the system parameter 'p' is –

a) $\sqrt{3}$ b) 1

c) $\frac{2}{\sqrt{3}}$ d) $\frac{\sqrt{3}}{2}$

**Answer**: (c)

**Frequency response analysis of second order system –**

Consider a second order system having transfer function –

$$T(s) = \frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

Where,

$\xi$ = damping factor

$\omega_n$ = undamped natural frequency

The sinusoidal transfer function of the system is –

$$T(j\omega) = \frac{\omega_n^2}{(j\omega)^2 + 2\xi(j\omega)\omega_n + \omega_n^2}$$

$$T(j\omega) = \frac{\omega_n^2}{\left(\omega_n^2 - \omega^2\right) + j2\xi\omega_n\omega}$$

$$T(j\omega) = \frac{1}{\left[1 - \left(\frac{\omega}{\omega_n}\right)^2\right] + 2j\xi\frac{\omega}{\omega_n}}$$

Put $u = \frac{\omega}{\omega_n}$ in above equation we get –

$$T(j\omega) = \frac{1}{\left(1 - u^2\right) + 2j\xi u}$$

- Magnitude of frequency response is known as magnitude response and angle is known as phase response.

Magnitude response is given by –

$$|T(j\omega)| = \frac{1}{\sqrt{\left(1 - u^2\right)^2 + 4\xi^2u^2}}$$

In decibel (dB), Magnitude (M) is given as

$$M_{dB} = 20\log_{10}|T(j\omega)|$$

And phase response is

$$\angle T(j\omega) = -\tan^{-1}\left(\frac{2\xi u}{1 - u^2}\right)$$

Or,

$$\angle T(j\omega) = -\tan^{-1}\left[\frac{2\xi\frac{\omega}{\omega_n}}{1 - \left(\frac{\omega}{\omega_n}\right)^2}\right]$$

**Case – 1.** At very low frequencies-

$$\frac{\omega}{\omega_n} << 1$$

$$\therefore |\, T(j\omega)\,| = 1$$

and $\quad \angle T(j\omega) = 0^\circ$

In decibel, $20\log |\, T(j\omega)\,| = 0\text{dB}$

**Case – 2.** At very high frequencies-

$$\frac{\omega}{\omega_n} >> 1$$

$$|\, T(j\omega)\,| = \frac{1}{\sqrt{\left(\frac{\omega}{\omega_n}\right)^4}} = \frac{1}{\left(\frac{\omega}{\omega_n}\right)^2}$$

In decibel, $20\log |\, T(j\omega)\,| = -40\log\frac{\omega}{\omega_n}$

$$20\log |\, T(j\omega)\,| = -40\log\omega + 40\log\omega_n$$

and $\quad \angle T(j\omega) = -\pi$

**Case – 3.** At some particular frequency-

$$\omega = \omega_n$$

$$|\, T(j\omega)\,| = \frac{1}{2\xi}$$

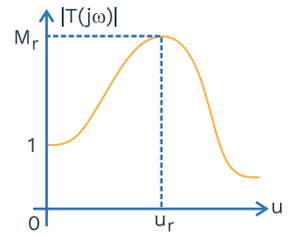
In decibel, $20\log |\, T(j\omega)\,| = -20\log 2\xi$

and $\quad \angle T(j\omega) = -\frac{\pi}{2}$

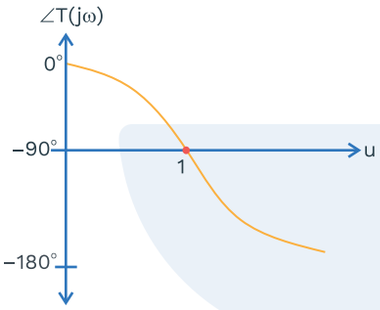
The steady state output of the system for a sinusoidal input with variable frequency $\omega$ is given by –

$$\because c(t) = \frac{1}{\sqrt{\left(1-u^2\right)^2 + (2\xi u)^2}} \sin\left(\omega t - \tan^{-1}\frac{2\xi u}{1-u^2}\right)$$

Frequency response magnitude and phase angle characteristics can be drawn as –




**Magnitude Response characteristic**




**Phase response characteristic**

## Frequency Domain Specification –

### 1. Resonant frequency ($\omega_r$) –

- It is the frequency at which the peak value of magnitude response occurs.
- Large value of resonant frequency indicates that the time response is fast.
- It is denoted by $\omega_r$.

At resonant frequency ($\omega_r$) –

$$\frac{d}{du}|T(j\omega)| = 0$$

$$\frac{d}{du}\left[\frac{1}{\sqrt{\left(1-u^2\right)^2+4\xi^2u^2}}\right] = 0$$

On solving this, we get –

$$u_r = \sqrt{1-2\xi^2} \quad \Rightarrow \omega_r = \omega_n\sqrt{1-2\xi^2}$$

Since, frequency is a real quantity, above expression is meaningful only for

$$1-2\xi^2 > 0$$

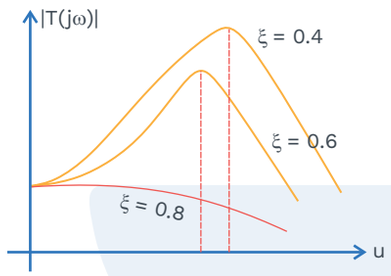
$$\xi < \frac{1}{\sqrt{2}}$$

So that, we can conclude that-

$0 < \xi < \frac{1}{\sqrt{2}}$, resonant peak occurs.

And, $\xi > \frac{1}{\sqrt{2}}$, no resonant peak occurs.

The graph of damping factor is –




**2. Resonant peak ($M_r$) –**

- Maximum value of the magnitude response is known as the resonant peak.
- It is denoted by $M_r$.
- It gives the information about the relative stability of the system.
- A large value of resonant peak implies an undesirable transient response.
- Resonant peak can be expressed as-

$$M_r = |T(j\omega_r)|$$

And $$|T(j\omega)| = \frac{1}{\sqrt{\left[1-\left(\frac{\omega}{\omega_n}\right)^2\right]^2 + 4\xi^2\left(\frac{\omega}{\omega_n}\right)^2}}$$

**Gray Matter Alert!!!**

For $0 < \xi < \frac{1}{\sqrt{2}}$, the resonant frequency always has a value less than natural frequency($\omega_n$) and the resonant peak has a value greater than 1.

Put the value of $\omega_r = \omega_n\sqrt{1-2\xi^2}$ in above expression, we get -

$$M_r = \frac{1}{2\xi\sqrt{1-\xi^2}}$$

The above expression is valid only when $\xi \leq \frac{1}{\sqrt{2}}$

**3. Bandwidth (dB) –**

- The bandwidth of a control system indicates the noise characteristic of a system and also gives a measure of the transient response properties.
- Bandwidth is defined as the range of frequencies in which the magnitude of closed loop does not drop to 70.7% or 3dB down from its zero frequency value.

**Definitions**

The range of frequencies occupied by a single channel or signal is called as bandwidth (BW).

**4. Cut-off frequency –**

- The frequency at which the magnitude of $|T(j\omega)|$ is $\frac{1}{\sqrt{2}}$ times of its maximum value is called as cut-off frequency.
- Cut-off frequency is denoted by $\omega_c$.

Magnitude is given as -

$$M = \frac{1}{\sqrt{\left(1-u^2\right)^2 + \left(2\xi u\right)^2}}$$

The normalized bandwidth $u_b = \frac{\omega_b}{\omega_n}$ of the second order system can be determined as follows –

$$M = \frac{1}{\sqrt{\left(1-u_b{}^2\right)^2 + 4\xi^2 u_b{}^2}} = \frac{1}{\sqrt{2}}$$

$$u_b^2 = \frac{-\left(4\xi^2 - 2\right) \pm \sqrt{\left(4\xi^2 - 2\right)^2 + 4}}{2}$$

On solving we get –

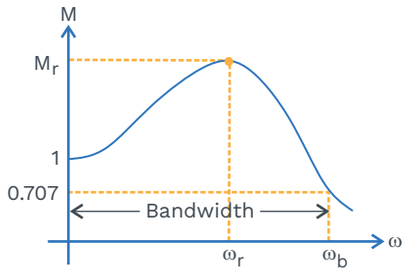
But,

$$u_b = \frac{\omega_b}{\omega_n}$$

$$\therefore \quad \omega_b = u_b \omega_n$$

Denormalized Bandwidth (BW)

$$\omega_b = \omega_n \left[\left(1-2\xi^2\right) + \sqrt{4\xi^4 - 4\xi^2 + 2}\right]^{1/2}$$




Approximately, we can write in linear form as –

$$\omega_b = -1.19\xi + 1.85$$

**5. Cut–off rate –**

- The cut-off rate is stated as the slope of the frequency response near the cut-off frequency.
- Cut-off rate is the rate of change of slope of the magnitude at cut off frequency.

**Example – 2**

If $M(j\omega) = \dfrac{100}{100 - \omega^2 + j10\sqrt{2}\omega}$, calculate $\omega_r$ and $M_r$.

**Solution:**

Given,

$$M(j\omega) = \frac{100}{100 - \omega^2 + j10\sqrt{2}\omega}$$

Compare with the sinusoidal transfer function, $T(j\omega) = \dfrac{\omega_n^2}{\omega_n^2 - \omega^2 + 2j\xi\omega_n\omega}$

we get,

$$\omega_n^2 = 100, \quad \omega_n = 10 \text{rad / sec}$$

and $$2\xi\omega_n = 10\sqrt{2}$$

$$\xi = \frac{10\sqrt{2}}{20} = \frac{1}{\sqrt{2}}$$

$$\omega_r = \omega_n\sqrt{1 - 2\xi^2}$$

$$\omega_r = 10\sqrt{1 - 2 \times \frac{1}{2}}$$

$$\omega_r = 0 \text{ rad/sec}$$

and $M_r = \dfrac{1}{2\xi\sqrt{1-\xi^2}}$

$$M_r = \frac{1}{2 \times \frac{1}{\sqrt{2}}\sqrt{1-\frac{1}{2}}}$$

$$M_r = 1$$

**Example – 3**

Magnitude of the frequency response for a second order system is 5 at 0 rad/sec. The magnitude peak $\frac{10}{\sqrt{3}}$ at $\frac{5}{\sqrt{2}}$ rad/sec. Calculate the transfer function of the system.

**Solution:**

Given –

DC gain, K = 5

$$M_r = \frac{10}{\sqrt{3}}, \quad \omega_r = \frac{5}{\sqrt{2}}$$

We know that, $M_r = \dfrac{K}{2\xi\sqrt{1-\xi^2}}$

$$\frac{K}{2\xi\sqrt{1-\xi^2}} = \frac{10}{\sqrt{3}}$$

$$\frac{5}{2\xi\sqrt{1-\xi^2}} = \frac{10}{\sqrt{3}}$$

$$16\xi^4 - 16\xi^2 + 3 = 0$$

$$\xi^2 = 0.25, 0.866$$

$\xi^2 = 0.866$ is invalid because at this value there is no resonant peak.

So, $\xi = 0.5$ is valid

and resonance frequency is given as –

$$\omega_r = \omega_n\sqrt{1-2\xi^2}$$

$$\frac{5}{\sqrt{2}} = \omega_n\sqrt{1-2(0.5)^2}$$

$$\omega_n = 5 \text{ rad/sec}$$

**Previous Year's Questions**

A system with zero initial conditions has the closed loop transfer function $T(s) = \dfrac{s^2+4}{(s+1)(s+4)}$, the system output is zero at the frequency-

a) 0.5 rad/sec b) 1 rad/sec

c) 2 rad/sec d) 4 rad/sec

**Answer:** (c)

Transfer function of second order system can be written as –

$$T(s) = \frac{K\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

$$T(s) = \frac{5 \times 25}{s^2 + 2 \times 0.5 \times 5s + 25}$$

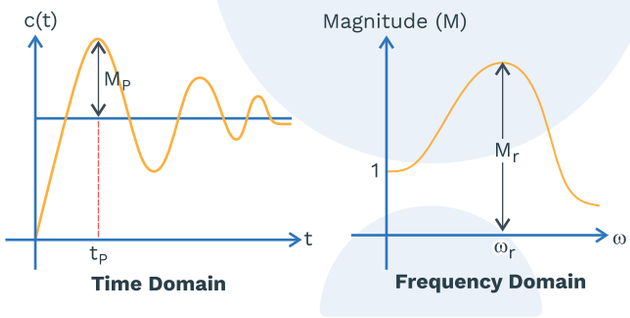
$$T(s) = \frac{125}{s^2 + 5s + 25}$$

**Rack your Brain**

What is use of frequency domain in control system?

**Relation between time and frequency response –**

- The correlation between time and frequency response exists only for the first and second order system.
- The transient response and the frequency response of linear systems are related through the Fourier integral.




- Fourier transform is a mapping between time domain and frequency domain.

**Stability from frequency response –**

Frequency response consist of two parts –

1. Magnitude response
2. Phase response

There are two types of stability i.e., absolute stability and relative stability.

- Absolute stability tells whether the system is stable or not and relative stability measures the degree of stability.
- To define the measure of relative stability, we use the concept of gain margin and phase margin.

**Gray Matter Alert!!!**

In time domain, relative stability is measured by damping ratio and maximum overshoot, while in frequency domain, relative stability is measured by resonant peak ($M_r$).

**Gain Margin –**

- Gain margin is the factor by which the open loop transfer function is multiplied to make the closed loop system marginally stable.
- Gain margin can also be defined as the reciprocal of the gain at the frequency at which the phase angle becomes 180°.
- The frequency at which the phase angle is 180° is called the phase cross-over frequency ($\omega_{pc}$).
- Mathematically, gain margin is the reciprocal of the magnitude of the G(jω) H(jω) at phase cross over frequency.

Gain margin $(GM) = \dfrac{1}{\left|G(j\omega)H(j\omega)\right|_{\omega=\omega_{pc}}}$

At phase cross over frequency $(\omega = \omega_{pc})$ –

$$\angle G(j\omega)H(j\omega) = -180^\circ$$

If, $|\, G(j\omega)H(j\omega)\,|_{\omega=\omega_{pc}} = X$

then,

Gain Margin $(GM) = \dfrac{1}{X}$

Generally, gain margin is expressed in decibels as-

$$GM_{dB} = 20\log\left(\frac{1}{X}\right)$$

**Gain crossover frequency ($\omega_{gc}$) –**

- The gain cross over frequency is the frequency at which |G(jω)H(jω)|, the magnitude of the open loop transfer function is unity.
  Mathematically, at gain crossover frequency ($\omega_{gc}$)-

$$|\, G(j\omega)H(j\omega)\,| = 1$$

In decibel, $\left|G(j\omega)H(j\omega)\right|_{\omega=\omega_{gc}} = 0dB$

**Phase Margin –**

- It is the amount of phase added to an open loop transfer function of a feedback system such that the closed loop system becomes marginally stable.
- Phase margin is represented in degrees or radians.
  Mathematically, the phase margin (PM) is –

$$PM = 180° + \phi$$

Where, $\phi = \angle G(j\omega)H(j\omega)\big|_{\omega=\omega_{gc}}$

**Gain margin and phase margin for second order system –**

Consider open loop transfer function –

$$G(s) = \frac{\omega_n^2}{s\left(s + 2\xi\omega_n\right)}$$

Put s = jω,

$$G(j\omega) = \frac{\omega_n^2}{j\omega\left(j\omega + 2\xi\omega_n\right)}$$

$$G(j\omega) = \frac{\omega_n^2}{-\omega^2 + j2\xi\omega_n\omega} \quad \ldots. (I)$$

$$|\, G(j\omega)\,| = \frac{\omega_n^2}{\sqrt{\left(-\omega^2\right)^2 + \left(2\xi\omega\omega_n\right)^2}}$$

At gain crossover frequency $(\omega_{gc})$,

$$\left|G\left(j\omega_{gc}\right)\right| = 1$$

$$\frac{\omega_n^2}{\omega_{gc}\sqrt{\omega_{gc}^2 + 4\xi^2\omega_n^2}} = 1$$

On solving, we get –

$$\omega_{gc} = \omega_n\sqrt{-2\xi^2 + \sqrt{4\xi^4 + 1}} \ldots\ldots (II)$$

At phase cross over frequency $\left(\omega_{pc}\right)$ -

$$\angle G\left(j\omega_{pc}\right) = -180°$$

$$G(j\omega) = \frac{\omega_n^2}{-\omega^2 + j2\xi\omega_n\omega}$$

$$\angle G(j\omega) = -180° + \tan^{-1}\left(\frac{2\xi\omega_n}{\omega}\right)$$

Thus,

$$-180° + \tan^{-1}\left(\frac{2\xi\omega_n}{\omega_{pc}}\right) = -180°$$

$$\tan^{-1}\left(\frac{2\xi\omega_n}{\omega_{pc}}\right) = 0$$

On solving, $\omega_{pc} = \infty$ rad/sec.

**Gray Matter Alert!!!**

- If $\omega_{gc} < \omega_{pc}$ and both gain margin and phase margin is positive then the system is stable.
- If $\omega_{gc} > \omega_{pc}$ and both gain margin and phase margin is negative then the system is unstable.
- If $\omega_{gc} = \omega_{pc}$ then the system is marginally stable and gain margin and phase margin both are equal to zero.

$$GM = \frac{1}{|G(j\omega)|_{\omega_{pc}}} = \frac{\omega_{pc}\sqrt{\omega_{pc}^2 + 4\xi^2\omega_n^2}}{\omega_n^2}$$

$$GM = \infty$$

And, $$PM = 180^\circ + \angle G(j\omega)\Big|_{\omega=\omega_{gc}}$$

$$= 180^\circ - 180^\circ + \tan^{-1}\left(\frac{2\xi\omega_n}{\omega_{gc}}\right)$$

On solving,

$$PM = \tan^{-1}\left[\frac{2\xi}{\sqrt{-2\xi^2 + \sqrt{4\xi^4 + 1}}}\right]$$

**Example – 4**

The forward path transfer function of a unity feedback control system is –

$$G(s) = \frac{100}{s(s+6)}$$

Determine the resonant peak and bandwidth of the closed loop system.

**Solution:**

Given, $$G(s) = \frac{100}{s(s+6)}$$

and $$H(s) = 1$$

Closed loop transfer function is $$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

$$\frac{C(s)}{R(s)} = \frac{\frac{100}{s(s+6)}}{1 + \frac{100}{s(s+6)}}$$

$$\frac{C(s)}{R(s)} = \frac{100}{s^2 + 6s + 100}$$

Compare with standard second order transfer function –

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

we get –

$$\omega_n^2 = 100 \, \text{rad/sec}$$

$\omega_n = 10$ rad/sec

and $2\xi\omega_n = 6$

$$\xi = \frac{6}{2 \times 10} = 0.3$$

We Know,

$$M_r = \frac{1}{2\xi\sqrt{1-\xi^2}}$$

$$M_r = \frac{1}{2 \times 0.3\sqrt{1-(0.3)^2}}$$

$M_r = 1.747$

And bandwidth is –

$$B.W. = \omega_n\sqrt{1-2\xi^2+\left(2-4\xi^2+4\xi^4\right)^{1/2}}$$

$$B.W. = 10\sqrt{1-2(0.3)^2+\left[(2-4(0.3)^2+4(0.3)^4\right]^{1/2}} = 14.53 \text{rad/sec}$$

**Example – 5**

The open loop transfer function of a unity feedback system is $\frac{5}{(s+3)^4}$.

Then the gain margin of the system (in dB) is –

**Solution:**

Given open loop transfer function is –

$$G(s)H(s) = \frac{5}{(s+3)^4}$$

Put $s = j\omega$,

$$G(j\omega)H(j\omega) = \frac{5}{(3+j\omega)^4}$$

$$\angle G(j\omega)H(j\omega) = -4\tan^{-1}\frac{\omega}{3}$$

At $\omega = \omega_{pc}$-

$$\angle G(j\omega_{pc})H(j\omega_{pc}) = -180°$$

$$\Rightarrow -4\tan^{-1}\frac{\omega_{pc}}{3} = -180°$$

$$\omega_{pc} = 3 \text{rad/sec}$$

**Rack your Brain**

What is the disadvantage of polar plot?

$$|G(j\omega)H(j\omega)| = \frac{5}{\left(\sqrt{3^2+\omega^2}\right)^4}$$

$$|G(j\omega)H(j\omega)|_{\omega=\omega_{pc}} = \frac{5}{(\sqrt{9+9})^4}$$

$$|G(j\omega)H(j\omega)|_{\omega_{pc}} = 0.0154$$

In decibel, $GM = 20\log\frac{1}{|G(j\omega)H(j\omega)|_{\omega=\omega_{pc}}} = 20\log\left(\frac{1}{0.0154}\right)$

$$GM_{dB} = 36.24 \text{ dB}$$

**POLAR PLOT –**

- Polar plot is used for the frequency response characteristics of the system.
- Polar plot is a plot between the magnitude and phase angle of G(jω) on polar coordinates by varying frequency from zero to infinity.
- In polar plot, magnitude of G(jω) is plotted as the distance from the origin and phase angle is measured from positive real axis.
- Polar plot is also known as the semi-Nyquist plot.

**Procedure to sketch polar plot –**

**Step-1.** Determine open loop transfer function G(s) of the system.
**Step–2.** Put s = jω in the transfer function and obtain |G(jω)| and ∠G(jω).
**Step–3.** At ω = 0 and ω = ∞ calculate |G(jω)| and ∠G(jω).
**Step-4.** Separate real and imaginary parts of G(jω).
**Step-5.** Equate the real part of G(jω) to zero and find the intersection with imaginary axis and calculate the value of G(jω) at the point of intersection.
**Step–6.** Equate the imaginary part of G(jω) to zero and find the frequencies at which plot intersects with real axis and obtain the value of G(jω) at that frequencies.
**Step–7.** By using the above steps, draw the polar plot.

**(a) Type-0 system -**

Consider a type-0 system having transfer function,

$$G(s)H(s) = \frac{1}{1+sT}$$

**Step–1.** Put s = jω,

$$G(j\omega)H(j\omega) = \frac{1}{1+j\omega T}$$

**Gray Matter Alert!!!**

In polar plot, positive angle is measured in counter clockwise direction and negative phase angle is measured in clockwise direction from the positive real axis.

**Step–2.** Magnitude can be written as –

$$M = \left|G(j\omega)H(j\omega)\right| = \frac{1}{\sqrt{1+\omega^2T^2}}$$

and angle is $\phi = \angle G(j\omega)H(j\omega) = -\tan^{-1}\omega T$

**Step–3.** Calculate the value of M and $\phi$ at different value of frequencies –

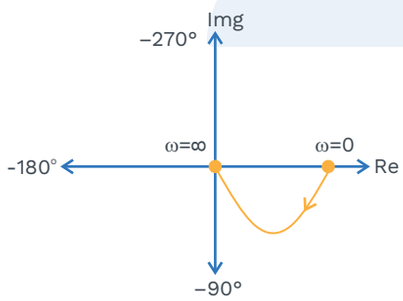
at $\omega = 0$, $M = 1$

and, $\phi = 0°$

Thus,

| **Frequency ($\omega$)(rad/sec)** | **Magnitude ($\|G(j\omega)H(j\omega)\|$)** | **Phase angle $\angle G(j\omega)H(j\omega)$ (degrees)** |
|---|---|---|
| 0 | 1 | 0° |
| $\frac{1}{T}$ | $\frac{1}{\sqrt{2}} = 0.707$ | -45° |
| $\infty$ | 0 | -90° |

The polar plot can be drawn as –




**(b) Type - 1 system –**

If $$G(s)H(s) = \frac{K}{s(1+sT_1)(1+sT_2)}$$

Put $s = j\omega$,

$$G(j\omega)H(j\omega) = \frac{K}{j\omega(1+j\omega T_1)(1+j\omega T_2)}$$

$$|G(j\omega)H(j\omega)| = \frac{K}{\omega\sqrt{1+\omega^2T_1^2}\sqrt{1+\omega^2T_2^2}}$$

$$\angle G(j\omega)H(j\omega) = -90° - \tan^{-1}\omega T_1 - \tan^{-1}\omega T_2$$

| Frequency (ω)(rad/sec) | Magnitude (\|G(jω)H(jω)\|) | Phase angle $\angle G(j\omega)H(j\omega)$ (degrees) |
|---|---|---|
| 0 | ∞ | -90° |
| ∞ | 0 | -270° |

$$G(j\omega)H(j\omega) = \frac{K}{j\omega(1+j\omega T_1)(1+j\omega T_2)}$$

On rationalizing and separating the real and imaginary parts –

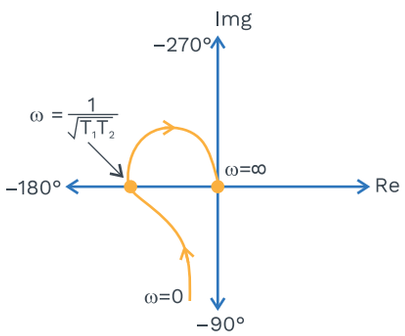
$$G(j\omega)H(j\omega) = \frac{-\omega K(T_1+T_2)}{\omega+\omega^3\left(T_1^2+T_2^2+\omega^2T_1^2T_2^2\right)} + \frac{j\left[K\omega^2T_1T_2-K\right)}{\omega+\omega^3\left(T_1^2+T_2^2+\omega^2T_1^2T_2^2\right)}$$

Equate imaginary part to zero –

$$\frac{K\omega^2T_1T_2-K}{\omega+\omega^3\left(T_1^2+T_2^2+\omega^2T_1^2T_2^2\right)} = 0$$

$$\therefore \quad \omega = \frac{1}{\sqrt{T_1T_2}}$$

The polar plot can be drawn as –




**(c) Type – 2 system –**

If $$G(s)H(s) = \frac{K}{s^2(1+sT_1)}$$

Put s = jω,

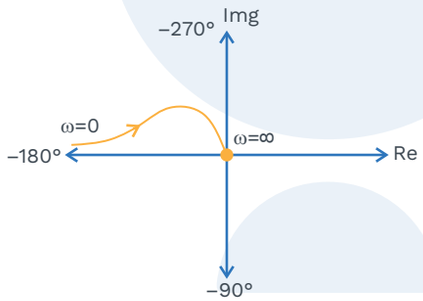
$$G(j\omega)H(j\omega) = \frac{K}{(j\omega)^2(1+j\omega T_1)}$$

$$|G(j\omega)H(j\omega)| = \frac{K}{\omega^2\sqrt{1+(\omega T_1)^2}}$$

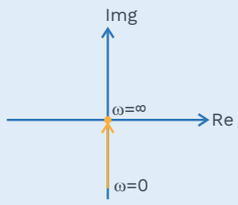
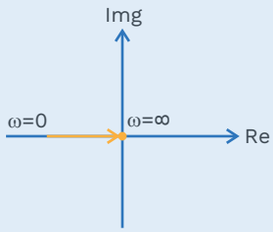
$$\angle G(j\omega)H(j\omega) = -90^\circ - 90^\circ - \tan^{-1}\omega T_1 = -180^\circ - \tan^{-1}\omega T_1$$

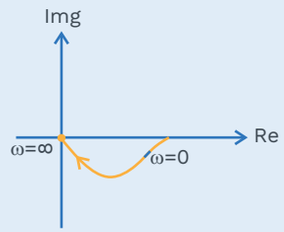
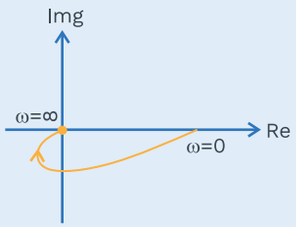
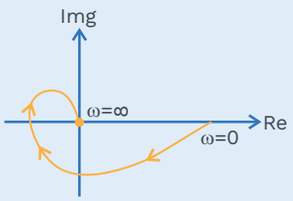
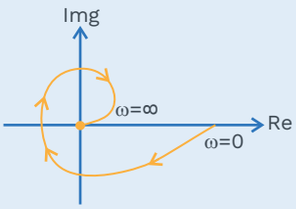
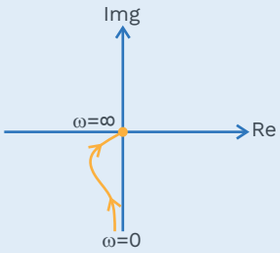
| Frequency ($\omega$)(rad/sec) | Magnitude ($\lvert G(j\omega)H(j\omega)\rvert$) | Phase angle $\angle G(j\omega)H(j\omega)$ (degrees) |
|---|---|---|
| 0 | $\infty$ | -180° |
| $\infty$ | 0 | -270° |

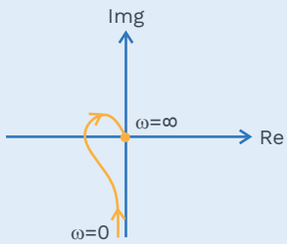
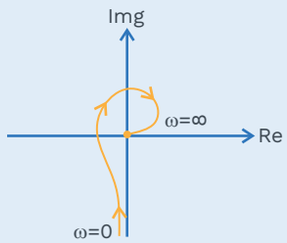
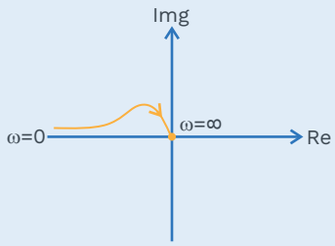
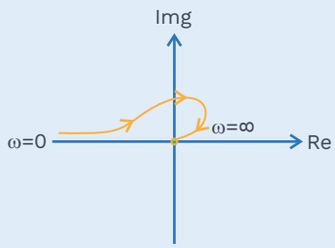
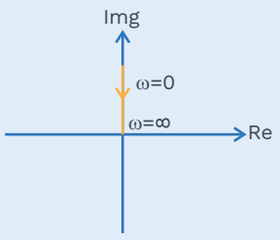
Polar plot is –

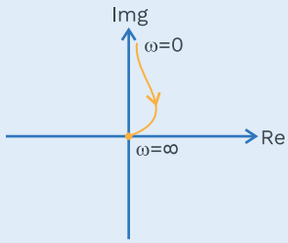



**Polar Plots of some standard function –**

| Function | Polar Plot |
|---|---|
| 1. $G(s) = \frac{1}{s}$ |   |
| 2. $G(s) = \frac{1}{s^2}$ |   |

| Function | Polar Plot |
|---|---|
| 3. $G(s) = \frac{1}{1+sT}$ |   |
| 4. $G(s) = \frac{1}{(1+sT_1)(1+sT_2)}$ |   |
| 5. $G(s) = \frac{1}{(1+sT_1)(1+sT_2)(1+sT_3)}$ |   |
| 6. $G(s) = \frac{1}{(1+sT_1)(1+sT_2)(1+sT_3)(1+sT_4)}$ |   |
| 7. $G(s) = \frac{1}{s(1+sT)}$ |   |

| Function | Polar Plot |
|---|---|
| 8. $G(s) = \dfrac{1}{s(1+sT_1)(1+sT_2)}$ |   |
| 9. $G(s) = \dfrac{1}{s(1+sT_1)(1+sT_2)(1+sT_3)}$ |   |
| 10. $G(s) = \dfrac{1}{s^2(1+sT)}$ |   |
| 11. $G(s) = \dfrac{1}{s^2(1+sT_1)(1+sT_2)}$ |   |
| 12. $G(s) = \dfrac{1}{s^3}$ |   |

| Function | Polar Plot |
|---|---|
| 13. $G(s) = \frac{1}{s^3(1+sT)}$ |  |



**Stability from polar plot –**

- Polar plots are used to determine the stability of the system.
- The stability of control system is measured from the two quantities which are gain margin and phase margin.
  The characteristic equation is –

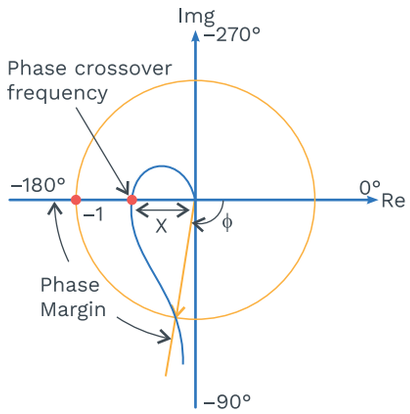
  $1 + G(s)H(s) = 0$

  $G(s)H(s) = -1$

  $G(s)H(s) = -1 + j0$

  $(-1 + j0)$ is a critical point.

  At critical point, magnitude = 1 and angle = ± 180°
- When the polar plot passes through the critical point $(-1 + j0)$, the gain margin becomes zero, this indicates that the gain cannot be increased because the system is at the verge of stability.
- The system is said to be stable, if the gain margin is positive. The negative gain margin indicates that the system is unstable.
- To check the stability of the system, draw the polar plot for the given transfer function.



- In the above figure, the polar plot intersects the negative real axis at a distance of X from the origin.
- The frequency where the polar plot crosses the negative real axis is called called as phase crossover frequency.
- If the critical point (–1 + j0) lies within the plot, it means the system is unstable.
- If the polar plot passes through the critical point (-1+j0), the system is marginally stable and if the critical point lies outside the polar plot, the system is stable.

**Gain Margin (GM) –**

- Gain margin is defined as how much the gain of a stable system increased at phase crossover frequency ($\omega_{pc}$) before the system becomes unstable.
- Gain margin is the reciprocal of the magnitude $|G(j\omega)|$ at the phase crossover frequency.

$$\text{Gain Margin (GM)} = \frac{1}{|\,G(j\omega)\,|_{\omega=\omega_{pc}}}$$

In decibel,

$$GM = -20 \log |G(j\omega_{pc})|$$

- If polar plot does not intersects the negative real axis-

$$|G(j\omega)| = 0$$

$$\Rightarrow \quad GM = \infty$$

- If the polar plot intersects the negative real axis between the origin and (–1 + j0) then,

$$GM > 0dB$$

- If the polar plot encloses the point (-1+j0) then,

$$|G(j\omega)| > 1$$

and, $GM < 0dB$

**Example – 6**

The transfer function of a system is $G(s)H(s) = \frac{k}{s(s+1)(s+5)}$. Draw the polar plot and also determine the range of k for stability.

**Solution:**

Given,

$$G(s)H(s) = \frac{k}{s(s+1)(s+5)}$$

Put $s = j\omega$,

**Gray Matter Alert!!!**

The value of $G(j\omega)H(j\omega)$ from $\omega = 0$ to $\omega = +\infty$ is also known as polar plot.

$$G(j\omega)H(j\omega) = \frac{k}{j\omega(1+j\omega)(5+j\omega)}$$

Magnitude of open loop transfer function is –

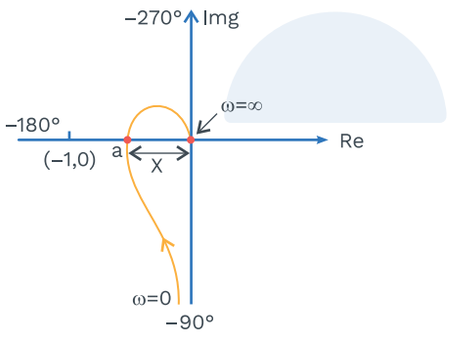
$$M = \frac{k}{\omega\sqrt{1+\omega^2}\sqrt{25+\omega^2}}$$

- Phase angle of the open loop transfer function is –

$$\phi = -90° - \tan^{-1}\omega - \tan^{-1}\frac{\omega}{5}$$

| **Frequency (ω)(rad/sec)** | **Magnitude (\|G(jω)H(jω)\|)** | **Phase angle $\angle G(j\omega)H(j\omega)$ (degrees)** |
|---|---|---|
| ω = 0 | ∞ | –90° |
| ω = ∞ | 0 | –270° |

The polar plot can be drawn as –




In the above polar plot, point 'a' is phase crossover point and frequency at this point 'a' is phase crossover frequency.
For stability, first find the value of X.
At point 'a', the phase angle is-

$$\phi = -180°$$

$$-180° = -90° - \tan^{-1}\omega - \tan^{-1} 0.2\omega$$

$$180°-90° = \tan^{-1}\omega + \tan^{-1} 0.2\omega$$

$$\tan 90° = \tan(\tan^{-1}\omega + \tan^{-1} 0.2\omega)$$

$$\infty = \frac{1.2\,\omega}{1-0.2\,\omega^2}$$

$1 - 0.2\,\omega^2 = 0$

$\omega = 2.236$ rad/sec

This is the phase crossover frequency ($\omega_{pc}$).

The magnitude of G(jω) at $\omega_{pc}$ is-

$$\left|G(j\omega)H(j\omega)\right|_{\omega=\omega_{pc}} = X = \frac{k}{2.236\sqrt{1+(2.236)^2}\,\sqrt{25+(2.236)^2}}$$

$$X = \frac{k}{30}$$

If,

k = 30, X = 1, system will be marginally stable.

k > 30, X > 1, system will be unstable.

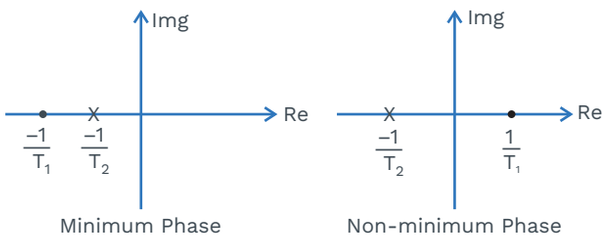
k < 30, X < 1, system will be stable.

**Minimum phase system**

- The transfer function having no poles and zeros in the right half of the s-plane are called is minimum phase system.
- The transfer function having poles or zeros in the right half of the s-plane is called as non-minimum phase system.

Consider, $G_1(s) = \dfrac{1+sT_1}{1+sT_2}$

and $G_2(s) = \dfrac{1-sT_1}{1+sT_2}$




- For minimum phase system, all poles and zeros must lie in the left half of s-plane.

**Inverse polar plot -**

- Inverse polar plot of G(jω) is $\dfrac{1}{G(j\omega)}$.

If $G(j\omega) = \dfrac{1}{j\omega}$

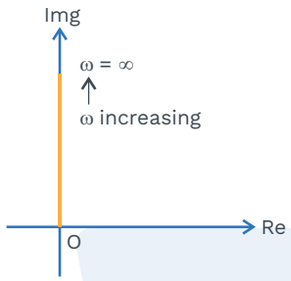
then,

$$G(j\omega)^{-1} = j\omega$$

Magnitude and phase angle can be written as –

$$M = \omega \text{ and } \phi = 90°$$

The inverse polar plot is –




**Example – 7**

If $G(s)H(s) = \dfrac{k}{(s+1)(1+4s)(1+3s)}$. Determine the value of phase crossover frequency.

**Solution:**

Given,

$$G(s)H(s) = \frac{k}{(s+1)(1+4s)(1+3s)}$$

Put $s = j\omega$,

$$G(j\omega)H(j\omega) = \frac{k+j0}{(1+j\omega)(1+4j\omega)(1+3j\omega)}$$

$$\angle G(j\omega)H(j\omega) = 0° - \left[\tan^{-1}\omega + \tan^{-1}4\omega + \tan^{-1}3\omega\right]$$

At $\omega = \omega_{pc}, \angle G(j\omega)H(j\omega) = -180°$

$$-\left[\tan^{-1}\omega_{pc} + \tan^{-1}4\omega_{pc} + \tan^{-1}3\omega_{pc}\right] = -180°$$

$$-\left[\tan^{-1}\omega_{pc} + \tan^{-1}\frac{7\omega_{pc}}{1-12\omega_{pc}^{\ 2}}\right] = -180°$$

$$\tan^{-1}\left[\frac{\omega_{pc} + \dfrac{7\omega_{pc}}{1-12\omega_{pc}^2}}{1-\dfrac{7\omega_{pc}^{\ 2}}{1-12\omega_{pc}^2}}\right] = 180°$$

$$\tan^{-1}\frac{\omega_{pc}\left(1-12\omega_{pc}^2\right)+7\omega_{pc}}{1-12\omega_{pc}-7\omega_{pc}^2}=180^\circ$$

$$\omega_{pc}\left(1-12\omega_{pc}^2+7\right)=0$$

$$\omega_{pc}\left(8-12\omega_{pc}^2\right)=0$$

$$\omega_{pc}=0 \text{ and } \omega_{pc}=\pm 0.816$$

For phase crossover frequency, only positive value is considered.

$$\therefore \quad \omega_{pc}=0.816\,\text{rad/sec}$$

## NYQUIST PLOT-

- Nyquist plots are used to draw the complete frequency response of the open loop transfer function.
- Nyquist plots are the continuation of polar plots for finding the stability of the closed loop control system by varying frequency ($\omega$) from $-\infty$ to $\infty$.

**Nyquist stability criterion-**

- Nyquist stability criteria gives the information about the stability and the relative stability of a system.
- Gain margin and phase margin are the two parameters by which we can measure the relative stability of a system.
- Nyquist stability criterion works on the 'principle of argument'.
- Principle of argument is related with the theory of mapping.

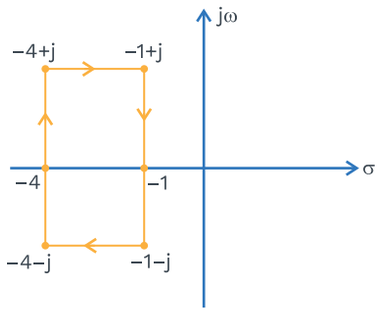
**Mapping of s-plane–**

- A set of values of 's' in s-plane are mapped to G(s)H(s) plane.
- For every value of 's' in s-plane can be mapped in G(s)H(s) plane. Consider a transfer function,

$$G(s)H(s)=\frac{1}{s+2}$$

- Any point in the s-plane can be mapped into G(s)H(s) plane, by putting the values of points in the transfer function.

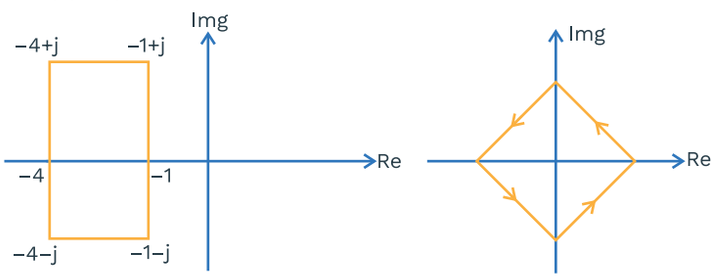
Let us consider some points on the s-plane as shown below–




Now, find all the values of the points of G(s)H(s) by putting the point in the transfer function –

| s | G(s)H(s) |
|---|---|
| –4 | -0.5 |
| -4+j | -0.4-j0.2 |
| -1+j | -0.5-j0.5 |
| -1 | 1 |
| -1-j | 0.5+j0.5 |
| -4-j | -0.4+j0.2 |

The mapping can be shown as –




- Every point in s-plane maps into one and only one point in the G(s)H(s) plane.

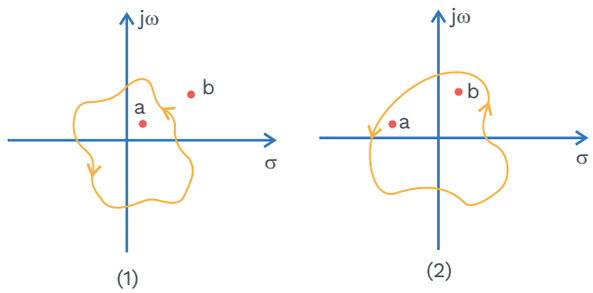
**Enclosement and Encirclement-**

- A point is said to be enclosed by a contour, if it lies to the right side of the path while travelling in the prescribed direction.

**Gray Matter Alert!!!**

Any close contour in the s-plane maps in the closed contour in the G(s)H(s) plane also.

- When a point or region in a complex function plane lies inside the closed contour, then it is said to be encircled by that contour.
  For Example-




In the above figure (1), point 'a' is encircled once in an anticlockwise direction while point 'b' is not encircled.
In figure (2), point 'a' and point 'b' both are enclosed by the contour.

**Principle of Argument:**

It states that if there are P poles and Z zeros, enclosed by the closed path in s-plane, then the corresponding G(s)H(s) plane must encircle the origin (P–Z) times, thus the number of encirclements N is –

$$N = P - Z$$

Where, N = Number of encirclement of origin in G(s)H(s) plane.
P = Number of poles of G(s)H(s) lying in right half side of s-plane.
Z = Number of zeros of G(s)H(s) lying in right half side of s-plane.

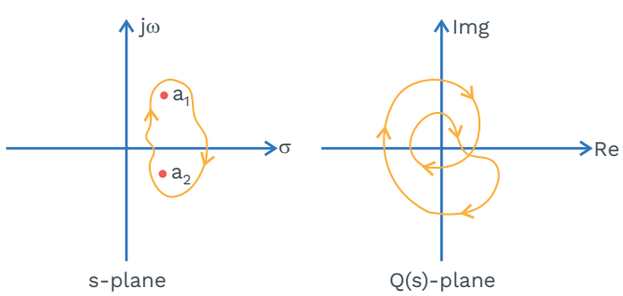
- "The relation between enclosure of poles and zeros by the contour in s-plane and encirclement of origin by the contour in G(s)H(s) plane is known as principle of argument."

**Rack your Brain**

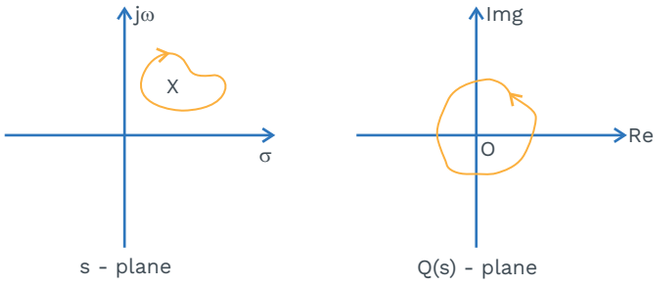
Why do we need to define contour in Nyquist stability criterion?

**Consider the following example –**

1. If the s-plane contour encloses two zeros, then the Q(s) plane contour encircles the origin twice in the clockwise direction as shown below -



2. If the contour in s-plane encircle 'P' number of poles in clockwise direction then the corresponding Q(s) contour will encircle the origin 'P' times in anticlockwise direction.




- If the s-plane contour encircles both the zeros and poles in clockwise direction then the corresponding G(s)H(s) contour encircles the origin of Q(s) plane N times i.e. (P–Z) in a clockwise direction.

**Gray Matter Alert!!!**

When Nyquist contour is mapped to G(s)H(s) plane then it becomes Nyquist plot.

**Nyquist stability criterion-**

- Nyquist stability criterion proclaims that the number of encirclements about the critical point (1 + j0) must be equal to the poles of the characteristic equation of an open loop transfer function in the right half of the 's' plane. The shift in origin to (1 + j0) gives the characteristic equation of s-plane.
- Consider the characteristic equation of the system -

$$Q(s) = 1 + G(s)H(s) = 0 \ldots (I)$$

**Definitions**

Nyquist criterion is a graphical method of determining stability of feedback control system by using the Nyquist plot of their open loop transfer functions.

Standard pole-zero form of the open loop transfer function can be written as –

$$G(s)H(s) = \frac{K(s+Z_1)(s+Z_2).....(s+Z_m)}{(s+P_1)(s+P_2).....(s+P_n)}, \ m \le n$$

From equation (I) –

$$Q(s) = 1 + \frac{K(s+Z_1)(s+Z_2).....(s+Z_m)}{(s+P_1)(s+P_2).....(s+P_n)}$$

$$Q(s) = \frac{(s+P_1)(s+P_2).....(s+P_n) + K(s+Z_1)(s+Z_2).....(s+Z_m)}{(s+P_1)(s+P_2).....(s+P_n)}$$

$$Q(s) = \frac{(s+Z_1')(s+Z_2').....(s+Z_m')}{(s+P_1)(s+P_2).....(s+P_n)} \quad ..... (II)$$

- From the equation (II), we can say that the poles of Q(s) are at $-P_1$, $-P_2$, ..... $-P_n$ are same as the open loop poles of the system.
- We know that the closed loop control system is stable, if all the poles of the closed loop transfer function are in left half of the s-plane. So, the poles of the closed loop transfer function are nothing but the roots of the characteristic equation.
- Let us consider that there are 'Z' zeros and 'P' poles of Q(s) in the right half of s-plane. Then,

  $N = P - Z$

  For the system to be stable, there should be no zeros of Q(s) in the right half of s-plane i.e.,

  $Z = 0$

  This condition is possible if,

  $N = P$

  Where,

  P = Total number of poles of the characteristic equation lying in the right half of s-plane or poles of open loop transfer function

  Z = Zeros of the characteristic equation lying in the right half of s-plane or poles of the closed loop transfer function.

  N = Number of encirclement around (–1, j0) for closed loop transfer function.

  N > 0, Anticlockwise direction (ACW)

  N< 0, Clockwise direction (CW)

  This means that the closed loop system is stable if the number of counter-clockwise encirclements of the origin of the Q(s) plane is equal to the number of poles on the right half of s-plane of Q(s) of open loop transfer function G(s)H(s).

**Stability using Nyquist plot:**

- From the Nyquist plot, we can determine both stability and the relative stability of a closed loop control system.
- Nyquist criteria is applicable for all types of system i.e., minimum phase system and non-minimum phase system.

**Gray Matter Alert!!!**

- The poles of the characteristic equation are same as that of the poles of the open loop transfer function.
- The zeros of the characteristic equation are same as that of the poles of the closed loop transfer function.

- There are mainly two criteria for stability of feedback system.

**Criteria -1:**

$$N = P - Z$$

N = Number of encirclement of (0, 0) point
Where,
N > 0, anticlockwise direction
N < 0, clockwise direction
P = Number of open loop poles in right half of s-plane
Z = Number of open loop zeros in right half of s-plane

**Criteria – 2:**

$$N = P - Z$$

N = Number of encirclement of (–1, 0) point
P = Number of open loop poles in right half of s-plane
Z = Number of closed loop poles in right half of s-plane

- For closed loop stability, Z = 0

We can conclude the following cases –

**Case-1:** N >0 (anticlockwise encirclement of (–1, 0) point)
If Z = 0, then for closed loop stability the number of open loop poles in right half of s-plane should be equal to number of encirclement of (–1, 0) point.

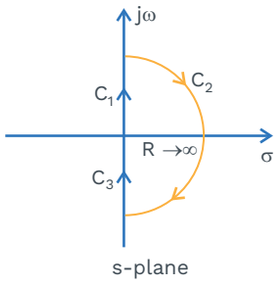
**Case-2:** N = 0 (No encirclement of (–1, 0) point) i.e., Z = P
Closed loop system will be stable if P = 0 or no open loop poles lie in right half of s-plane.

**Case-3:** N < 0 (clockwise encirclement (–1, 0) point) i.e., Z > P
Then the closed loop system will be unstable.

**Nyquist Contour –**
In order to find out zeros of Q(s) in the right half of s-plane, we choose a contour in such a way that completely encloses the right half of s-plane.

**Definitions**

Nyquist contour is a closed path in s-plane that encloses the entire right half of s-plane.



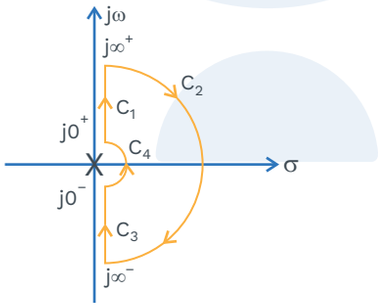

The Nyquist contour shown in above figure which is directed clockwise and comprises of an infinite line segment along the jω axis and an arc $C_2$ of infinite radius.

**Gray Matter Alert!!!**

Nyquist contour should never pass through any pole of G(s)H(s).

**Mapping of Nyquist Contour –**

When the transfer function G(s)H(s) has poles or zeros at the origin then the Nyquist contour consist of a semicircle near the origin with infinite radius which is as shown below –




$C_1$ : s = jω, where ω varies from 0 to ∞. $C_1$ represents the polar plot of the transfer function.

$C_2$ : $s = \lim_{R \to \infty} Re^{j\theta}$, where θ varies from +90° to –90° through origin.

$C_3$ : s = jω, where ω varies from –∞ to 0. $C_3$ represents the inverse polar plot of the transfer function.

$C_4$ : $s = \lim_{R \to 0} Re^{j\theta}$, where θ varies from –90° to 90° through the origin.

**Example - 8**

Consider $G(s)\,H(s)=\frac{1}{s^2}$. Sketch the complete Nyquist plot.

**Solution:**

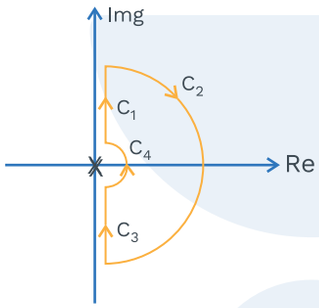
Given,

$$G(s)H(s)=\frac{1}{s^2}$$

Put $s = j\omega$,

$$G(j\omega)H(j\omega)=\frac{-1}{\omega^2}$$

The Nyquist path in s-plane for the given transfer function is-




$C_1$ : It is polar plot of G(s)H(s)

$$G(s)H(s)=\frac{-1}{\omega^2}$$

Magnitude, $M=\frac{1}{\omega^2}$

And phase, $\phi = -90° - 90° = -180°$

$C_2$ : $s=\lim_{R\to\infty} Re^{j\theta}$, $\theta \to 90°$ to $-90°$

$$G(s)H(s)=\frac{1}{s^2}=\lim_{R\to\infty}\frac{1}{R^2e^{j2\theta}}$$

$$G(s)H(s)=\lim_{R\to\infty}\frac{1}{R^2}e^{-j2\theta}$$

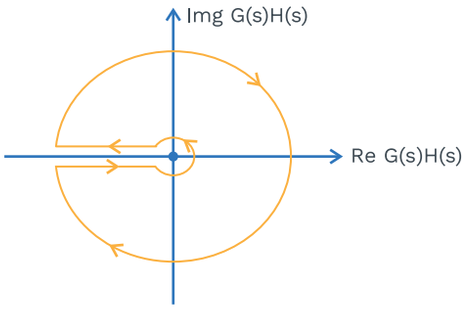
$C_3$ : It is an inverse polar plot.

$s = -j\omega$

$C_4$: $s=\lim_{R\to 0} Re^{j\theta}$, $\theta \to -90°$ to $90°$

$$G(s)H(s)=\lim_{R\to 0}\frac{1}{R^2}e^{-j2\theta}$$

This is the equation of semicircle with radius of ∞ from 180° to –180° in clockwise direction.
The shape of Nyquist plot can be drawn as -




Encirclement of origin (0, 0) is-

$$N = 1 - 1 = 0$$

**Example-9**

Draw the Nyquist plot to determine the phase crossover frequency and gain margin for the given open loop transfer function (If K = 2)-

$$G(s)\,H(s) = \frac{2.5K}{s(1+0.4\,s)(1+0.2s)}$$

**Solution:**
Given,

$$G(s)\,H(s) = \frac{2.5K}{s(1+0.4\,s)(1+0.2s)}$$

Put $s = j\omega$,

$$G(j\omega)\,H(j\omega) = \frac{2.5K}{j\omega(1+0.4\,j\omega)(1+0.2j\omega)}$$

Magnitude, $M = \left|G(j\omega)\,H(j\omega)\right| = \frac{2.5K}{\omega\sqrt{1+(0.4\,\omega)^2}\,\sqrt{1+(0.2\omega)^2}}$

And phase, $\phi = \angle G(j\omega)\,H(j\omega) = -90° - \tan^{-1}0.4\omega - \tan^{-1}0.2\omega$

$$\phi = -90° - \tan^{-1}\frac{0.6\,\omega}{1-0.08\,\omega^2}$$

1. When $\omega \to 0$,

$$M = \left|G(j0)\,H(j0)\right| \to \infty$$

$$\phi = \angle G(j0)\,H(j0) \to -90°$$

2. When $\omega \to \infty$,

$$M = \left| G(j\infty)\, H(j\infty) \right| \to 0$$

$$\phi = \angle G(j\infty)\, H(j\infty) \to -270°$$

3. Nyquist plot crosses the negative real axis at the phase crossover frequency $\omega_2$

$$\therefore \quad \angle G(j\omega_2)H(j\omega_2) = -180°$$

$$1 - 0.08\,\omega_2^{\,2} = 0$$

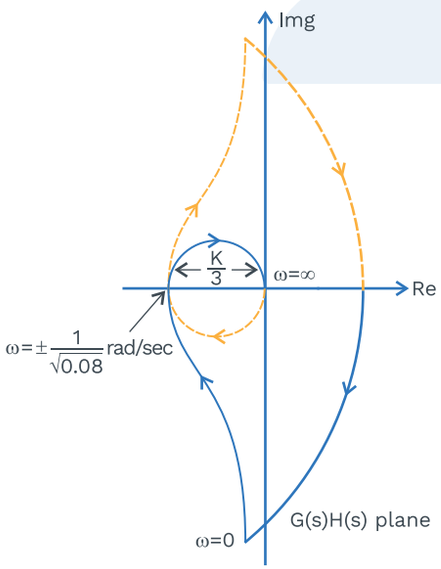
$$\omega_{pc} = \omega_2 = \frac{1}{\sqrt{0.08}} \text{ rad / sec}$$

Now,
$$\left|G(j\omega_2)\, H(j\omega_2)\right| = \frac{2.5K}{\omega_2\sqrt{1+\left(0.4\,\omega_2\right)^2}\sqrt{1+\left(0.2\,\omega_2\right)^2}}$$

$$= \frac{2.5K}{\frac{1}{\sqrt{0.08}}\sqrt{1+\frac{0.16}{0.08}}\sqrt{1+\frac{0.04}{0.08}}} = \frac{K}{3}$$

Thus, the Nyquist plot crosses the negative real axis of G(s)H(s) plane at $\frac{-K}{3}$

The complete Nyquist plot is -




$$GM(dB) = 20 \log_{10} \frac{1}{(K/3)} = 20 \log_{10} \frac{3}{K}$$

$$GM(dB) = 20 \log_{10} 3 - 20 \log_{10} K$$

For K = 2,

$$GM(dB) = 20\log_{10} 3 - 20\log_{10} 2 = 3.52$$

**Relative Stability using Nyquist plot:**

- From the Nyquist plot we can determine both stability and the relative stability of a closed loop control system.
- Nyquist criteria is applicable for all types of system i.e., minimum phase system and non-minimum phase system.
- There are mainly two criteria for stability of feedback system.
- Relative stability of closed loop systems can be measured in terms of the two parameters i.e., gain margin and phase margin.

**Gray Matter Alert!!!**

- Polar plot maps imaginary axis of s-plane to G(s)H(s) plane.
- Nyquist plot maps a contour (path) in s-plane to G(s)H(s) plane.

**Gain Margin-**

- Gain margin is defined as the reciprocal of the gain at the frequency at which the phase angle becomes 180°.

$$GM = \frac{1}{X}$$

Where,

$$X = \left|G(j\omega)\, H(j\omega)\right|_{\omega=\omega_{pc}}$$

**Phase Crossover Frequency ($\omega_{pc}$) –**

The frequency at which the Nyquist plot intersects the negative real axis (phase angle is 180°) is called the phase crossover frequency.

**Gray Matter Alert!!!**

Gain margin and phase margin concepts are applicable to open loop systems only.

**Gain Crossover Frequency –**

The frequency at which the Nyquist plot is having the magnitude of one is known as the gain crossover frequency.

**Phase Margin -**

- It is the amount of phase added to an open loop transfer function of a feedback system such that closed loop system becomes marginally stable.
- Phase margin (PM) is equal to the sum of 180° and the phase angle at the gain crossover frequency.

$$PM = 180° + \phi$$

Where,

$$\phi = \angle G(j\omega)H(j\omega)\Big|_{\omega=\omega_{gc}}$$

The stability of a control system based on the relation between the gain margin and the phase margin which can be illustrated as below –

1. If the gain margin is greater than one and the phase margin is positive, then the control system is stable.
2. If the gain margin is less than one and the phase margin is negative then the control system is unstable.
3. If the gain margin is equal to one and phase margin is zero degrees, then the control system is marginally stable.

**Example – 10**

Using Nyquist criterion determine whether the closed loop system is stable or not, whose open loop transfer function is given as-

$$G(s)H(s) = \frac{60}{(s+1)(s+2)(s+5)}$$

**Solution:**

Given,

$$G(s)H(s) = \frac{60}{(s+1)(s+2)(s+5)}$$

Put $s = j\omega$,

$$G(j\omega)\,H(j\omega) = \frac{60}{(1+j\omega)(2+j\omega)(5+j\omega)}$$

Rationalizing the above equation and separating the real and imaginary parts, we get –

$$G(j\omega)\,H(j\omega) = \frac{60(10-8\omega^2)}{(1+\omega^2)(4+\omega^2)(25+\omega^2)} - j\frac{60(17\omega-\omega^2)}{(1+\omega^2)(4+\omega^2)(25+\omega^2)}$$

By equating real part to zero-

$$\frac{60(10-8\omega^2)}{(1+\omega^2)(4+\omega^2)(25+\omega^2)} = 0$$

$$8\omega^2 = 10$$

$$\omega = 1.11 \text{ rad/sec}$$

Point of intersection on imaginary axis $= \left| G(j\omega)H(j\omega)\right|_{\omega=1.11}$

$$\left| G(j\omega)H(j\omega)\right| = \frac{60}{\sqrt{1+\omega^2}\ \sqrt{4+\omega^2}\ \sqrt{25+\omega^2}}$$

$$\left|G(j\omega)H(j\omega)\right|_{\omega=1.11} = \frac{60}{1.5\times 2.29\times 5.12} = 3.40$$

By equating imaginary part to zero,

$$\frac{60(17\omega-\omega^3)}{(1+\omega^2)(4+\omega^2)(25+\omega^2)} = 0$$

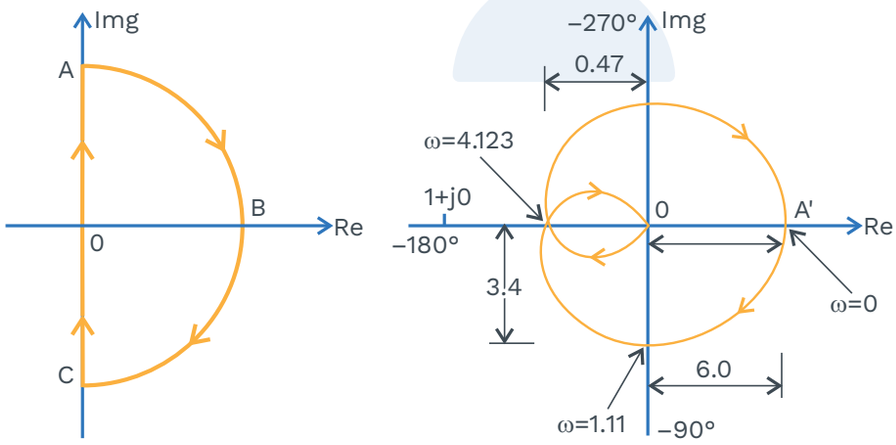
$\omega$ = 4.123 rad/sec.

Phase angle is–

$$\angle G(j\omega)\,H(j\omega) = -\tan^{-1}\omega - \tan^{-1}\frac{\omega}{2} - \tan^{-1}\frac{\omega}{5}$$

| **Frequency ($\omega$)(rad/sec)** | **Magnitude ($\lvert G(j\omega)H(j\omega)\rvert$)** | **Phase angle $\angle G(j\omega)H(j\omega)$ (degrees)** |
|---|---|---|
| 0 | 6 | 0° |
| $\infty$ | 0 | –270° |

Nyquist contour and Nyquist plot can be drawn respectively as –




∵ No open loop pole is in right half plane

∴ P = 0

The critical point (–1, 0) is not encircled by the plot.

∴ N = 0

N = P – Z

0 = 0 – Z

⇒ Z = 0

Thus, no closed loop pole is in the right half of s-plane, so the system is stable.

**Example-11**

Calculate the number of encirclements of point (0, 0) and (–1, 0) in complex plane by Nyquist plot of trasfer function,

$$G(s)H(s) = \frac{1-s}{6+3s}$$

**Solution:**

Case: 1 – For encirclement of origin (0, 0)

$$G(s)H(s) = \frac{1-s}{6+3s}$$

$s_P = -2$, $P = 0$ (no pole in right hand plane)
$s_Z = 1$, $Z = 1$ (one zero in right hand plane)
We know,

$$N = P - Z = 0 - 1$$

$$N = -1$$

$\therefore$ The number of encirclement of origin (0, 0) is one in clockwise direction.

Case: 2- For encirclement of (–1, 0)

$$G(s)H(s) = \frac{1-s}{6+3s}$$

$$1 + G(s)H(s) = 1 + \frac{1-s}{6+3s}$$

$$1 + G(s)H(s) = \frac{7+2s}{6+3s}$$

$s_P = -2$, $P = 0$ (no pole in right hand plane)
$s_Z = -3.5$, $Z = 0$ (no zero in right hand plane)
$N = P - Z = 0 - 0 = 0$

Thus, there is no encirclement of (–1, 0).

**Example – 12**

The gain margin of the system is –26dB. The point of intersection of Nyquist plot with respect to negative real axis is -

(a) 0.5 (b) 1/20
(c) 20 (d) 0.2

**Solution:**

Given,
Gain margin (GM) = –26dB

$$20 \log \frac{1}{|G(j\omega)|} = -26$$

$$\log \frac{1}{|G(j\omega)|} = -1.3$$

$$|G(j\omega)| = 10^{1.3} = 20$$

Gain at $\omega_{pc} = 20$

**Example –13**

The open loop transfer function of a unity feedback system is –

$$G(s) = \frac{1}{s(1+s)(1+2s)}$$

Draw the Nyquist plot of closed loop system and determine stability of the system.

**Solution:**

Given,

$$G(s) = \frac{1}{s(1+s)(1+2s)}$$

Put $s = j\omega$,

$$G(j\omega) = \frac{1}{j\omega(1+j\omega)(1+2j\omega)}$$

Rationalizing the above equation and separate the real and imaginary part –

$$G(j\omega) = \frac{-3}{\omega(1+4\omega^2)(1+\omega^2)} - j\frac{1-2\omega^2}{\omega(1+\omega^2)(1+4\omega^2)}$$

To get point of intersection on real axis, equate imaginary part to zero.

$$1 - 2\omega^2 = 0$$

$$\omega_{pc} = \frac{1}{\sqrt{2}} = 0.707$$

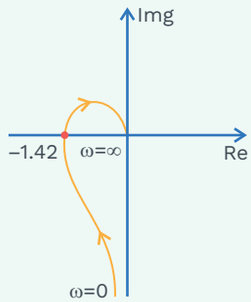
At $\omega_{pc}$ -

$$|G(j\omega)|_{\omega_{pc}} = 0.67$$

Complete Nyquist plot can be drawn as -

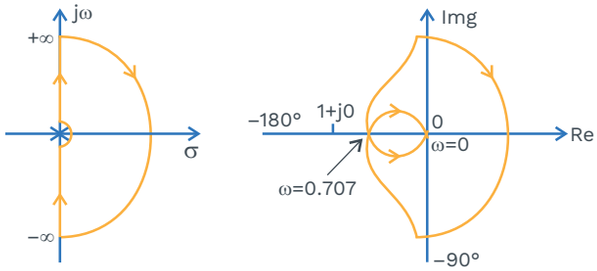
**Previous Year's Questions**

The polar plot of an open loop system is shown below. The closed loop system is -




(a) always stable
(b) marginally stable
(c) unstable with one pole on the right hand plane
(d) unstable with two poles on the right hand plane

**Answer:** (d)




The point (–1 + j0) is not encircled,

$\therefore$ N = 0

P = 0

N = P – Z

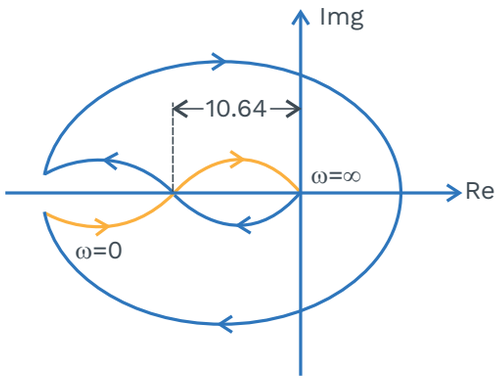
0 = 0 – Z

Z = 0

Thus, no closed loop poles lies in right half of s-plane.
$\therefore$ The system is stable.

**Example - 14**

The Nyquist plot of a unity feedback system having an open loop transfer function $G(s) = \frac{2s+1}{s^2(s+1)(2s+2)}$ is as shown below. Find the number of poles of the closed loop system that lies in right half of plane.




**Solution:**

Given,

$$G(s) = \frac{2s+1}{s^2(s+1)(2s+2)}$$

Open loop pole lies in right hand plane (P) = 0
From the Nyquist plot, the critical point (–1, 0) is encircled two times in a clockwise direction.

$$N = -2$$

$$N = P - Z$$

$$-2 = 0 - Z$$

$$Z = 2$$

Thus, two closed loop poles lies in the right half of s-plane.

## BODE PLOT-

- Bode plot is a logarithmic plot which consists of two plots –
  1. Magnitude plot
  2. Phase plot

**Rack your Brain**

Why bode plot is drawn on semi log paper?

- Bode plot is a graphical representation of the transfer function for determining the stability of the control system.
- Bode plots are plotted against frequency in logarithmic scale.
- The curves are generally drawn on a semi-log paper, using the log scale for frequency, linear scale for magnitude in decibel (dB) and phase angle in degrees.
  The magnitude of the open loop transfer function in dB is –

$$M = 20 \log |G(j\omega)\, H(j\omega)|$$

  And phase angle of the open loop transfer function in degree is -

$$\phi = \angle G(j\omega)\, H(j\omega)$$

**Bode plot for transfer function -**

Consider an open loop transfer function of a closed loop control system,

$$G(s)\, H(s) = \frac{K\left[(1+sT_1)(1+sT_2)....\right]\omega_n^{\,2}}{s^N\left[(1+sT_a)(1+sT_b)....\left(s^2+2\xi\omega_n s+\omega_n^{\,2}\right)\right]}$$

Put $s = j\omega$,

$$G(j\omega)\, H(j\omega) = \frac{K\left[(1+j\omega T_1)(1+j\omega T_2)....\right]\omega_n^{\,2}}{(j\omega)^N\left[(1+j\omega T_a)(1+j\omega T_b)....\left\{\left(\omega_n^2-\omega^2\right)+j2\xi\omega_n\omega\right\}\right]}$$

$$G(j\omega)\,H(j\omega) = \frac{K\left[\left(1+j\omega T_1\right)\left(1+j\omega T_2\right)\right]}{(j\omega)^N\left(1+j\omega T_a\right)\left(1+j\omega T_b\right).....\left[1+j2\xi\left(\frac{\omega}{\omega_n}\right)+\left(\frac{j\omega}{\omega_n}\right)^2\right]} \quad .... (I)$$

Constant multiplier K is given by –

$$K = \lim_{\omega \to 0} (j\omega)^N\, G(j\omega)\, H(j\omega)$$

From equation (I) log magnitude is given by –

$$20 \log \left|G(j\omega)\right| = 20 \log K + 20 \log |1+j\omega T_1| + 20 \log \left|1+j\omega T_2\right| + .....$$

$$-20N\log \omega - 20 \log \left|1+j\omega T_a\right| - 20 \log \left|1+j\omega T_b\right| .....$$

$$-20 \log \left|1+j2\xi\left(\omega/\omega_n\right)-\left(\omega/\omega_n\right)^2\right| .....$$

And phase angle is given by -

$$\angle G(j\omega)H(j\omega) = \tan^{-1}\omega T_1 + \tan^{-1}\omega T_2 + .....$$

$$-90°N - \tan^{-1}\omega T_a - \tan^{-1}\omega T_b ..... - \tan^{-1}\left(\frac{2\xi\omega\,\omega_n}{\omega_n^2-\omega^2}\right)$$

The basic factors that frequently occurs in a transfer function are –

1. Constant gain K
2. Poles at the origin $1/(j\omega)^N$
3. Zero on real axis $(1 + j\omega T)$
4. First order factor $(1 + j\omega T)^{\pm 1}$

5 Quadratic factors $\left[1+2\xi\left(\frac{j\omega}{\omega_n}\right)+\left(\frac{j\omega}{\omega_n}\right)^2\right]^{\pm 1}$

**1. Constant gain (K) -**

$G(s)H(s) = K$

Let us assume,

$H(s) = 1, \qquad G(s) = K$

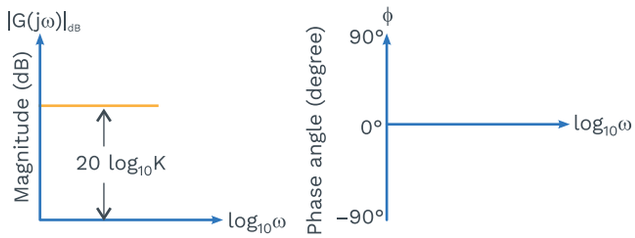
Put $s = j\omega$,

$$G(j\omega) = K$$

$$20 \log_{10} \left|G(j\omega)\right| = 20 \log_{10} K$$

Phase angle $(\phi) = \angle\, G\,(j\omega) = 0°$

Magnitude is independent of $\log_{10} \omega$ and phase angle is always zero. The plots can be drawn as -




**2. Zero at the origin -**

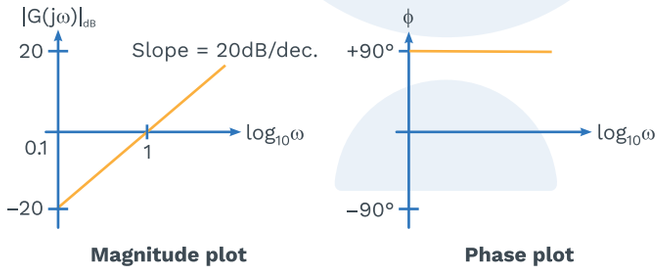
$G(s) = s$

$G(j\omega) = j\omega$

$\left|G(j\omega)\right|_{dB} = 20 \log \omega$

$\angle G(j\omega) = +90°$

The magnitude plot is a straight line having a slope of +20dB/decade and angular phase shift is +90° as given below -




Intersection of magnitude plot with 0dB axis -

$0 = 20 \log_{10} \omega$

$\omega = 1 \text{ rad / sec}$

**3. Pole at the origin -**

$$G(s) = \frac{1}{s}$$

Put $s = j\omega$,

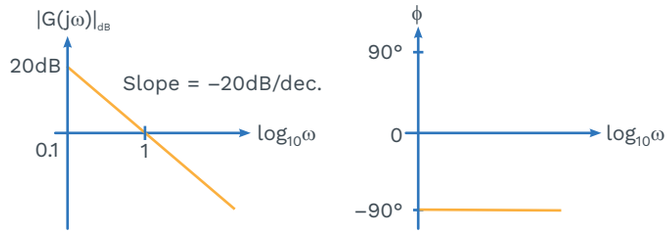
$$G(j\omega) = \frac{1}{j\omega}$$

$$\left|G(j\omega)\right| = \frac{1}{\omega}$$

$$\left|G(j\omega)\right|_{dB} = 20 \log \frac{1}{\omega}$$

$|G(j\omega)|_{dB} = -20 \log \omega$

$\angle G(j\omega) = -90°$




**Magnitude plot** **Phase plot**

**4. Multiple poles at origin -**

$$G(s) = \frac{1}{s^N}$$

Put s = jω,

$$G(j\omega) = \frac{1}{(j\omega)^N}$$

Magnitude in decibel (dB) –

$$20 \log |G(j\omega)| = 20 \log_{10} \frac{1}{\omega^N}$$

$$20 \log|G(j\omega)| = 20 \log_{10} (\omega)^{-N}$$

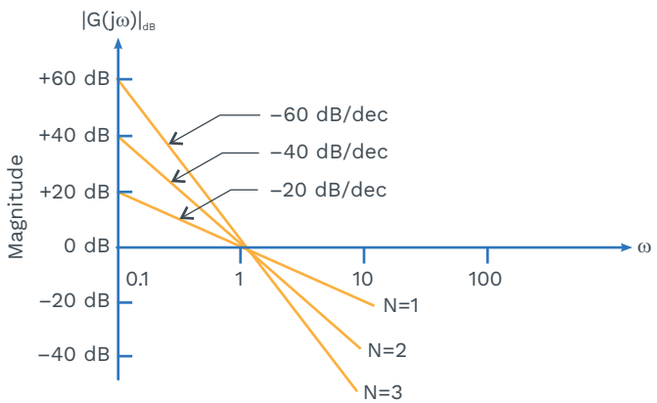
$$20 \log|G(j\omega)| = -20 N \log_{10} \omega$$

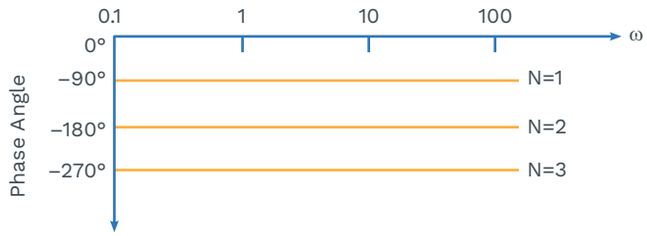
and $\angle G(j\omega) = -90° N$, N = 1, 2, 3, .....

**Gray Matter Alert!!!**

Bode plot shows both low and high frequency characteristics of the transfer function.

The magnitude plot is a straight line. For N = 1, the line has a slope of –20 dB/decade and angle –90°. For N = 2, the slope of the line will be –40 dB/decade and angle will be –180° and so on.






**Bode plot for the term $1/(j\omega)^N$**

**5. First order zero -**

$$G(s) = \left(1 + \frac{s}{\omega_c}\right)$$

where, $\omega_c$= corner frequency

Put s = jω,

$$G(s) = 1 + \frac{j\omega}{\omega_c}$$

Magnitude, $\left|G(j\omega)\right| = \sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^2}$

Magnitude in dB -

$$M = 20 \log \left|G(j\omega)\right| = 20 \log_{10} \sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^2}$$

We must take asymptotic approximation.

**Case: 1- Low frequency region ($\omega << \omega_c$) –**

$$0 < \frac{\omega}{\omega_c} << 1;\ M = 20 \log_{10} \sqrt{1} = 0\,\text{dB}$$

**Case: 2 – High frequency region –**

$$\frac{\omega}{\omega_c} >> 1$$

$$M = 20 \log \sqrt{\left(\frac{\omega}{\omega_c}\right)^2} = 20 \log \frac{\omega}{\omega_c}$$

$$M = 20 \log \omega - 20 \log \omega_c$$

Phase angle, $\phi = \angle G(j\omega) = \tan^{-1} \frac{\omega}{\omega_c}$

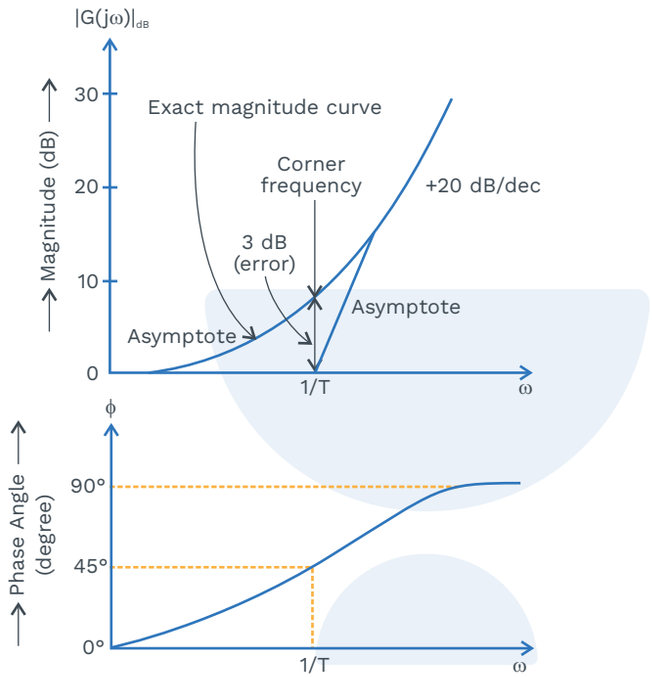
$\omega << \omega_c$, $\phi$ = 0°

$\omega >> \omega_c$, $\phi$ = 90°

**Gray Matter Alert!!!**

The frequency at which the slope of the magnitude plot changes is referred to as corner frequency.

Thus, the frequency response curve can be approximated by two straight lines, one is at 0dB slope and the other having the slope of the 20dB/decade. The exact and approximated curve of both magnitude and phase angle are given below –




**Procedure for drawing the bode plots –**

There are some following steps while constructing a bode plot –

1. Rewrite the given transfer function in the standard time constant form.
2. Identify the corner frequencies associated with each factor of the transfer function and arrange them in ascending order.
3. The approximate transfer function between every two-corner frequency as –

   If $\omega << \omega_c$, ignore $\frac{s}{\omega_c}$ term.

   $\omega >> \omega_c$, ignore '1'
4. Draw the asymptotic magnitude plot. The slope will change at each corner frequency by 20 dB/decade for zero and –20dB/decade for pole. For complex conjugate zero or pole, the slope will change by ±40dB/decade.

5. Draw the magnitude plots for each term and combine these plots properly.
6. Draw the phase angle curve for each factor and combine these plots properly.

**Example – 15**

The open loop transfer function of a unity feedback system is

$$G(s)H(s) = \frac{5(s+20)(s+50)}{(s+10)\,(s+100)}$$

Draw the bode magnitude plot.

**Solution:**

Given,

$$G(s)H(s) = \frac{5(s+20)(s+50)}{(s+10)\,(s+100)}$$

$$G(s)H(s) = \frac{5\times 20\times 50\left(1+\frac{s}{20}\right)\left(1+\frac{s}{50}\right)}{10\times 100\left(1+\frac{s}{10}\right)\left(1+\frac{s}{100}\right)}$$

$$G(s)H(s) = \frac{5\left(1+\frac{s}{20}\right)\left(1+\frac{s}{50}\right)}{\left(1+\frac{s}{10}\right)\left(1+\frac{s}{100}\right)}$$

DC gain, K = 5

Corner frequencies are 10, 20, 50, 100

Case: 1 If $\omega < 10$

then, $\omega$ is less than all corner frequencies.

So, that we can neglect $\frac{s}{\omega_c}$ in each term.

$\therefore$ G(s)H(s) = 5

Case: 2 If $10 < \omega < 20$

then, $G(s)H(s) = \frac{5\times 1\times 1}{\frac{s}{10}\times 1} = \frac{50}{s} \Rightarrow G(j\omega)H(j\omega) = \frac{50}{j\omega}$

Magnitude in dB, (M) = $20\log_{10}50 - 20\log_{10}\omega$

Slope of line = –20 dB/decade

Case: 3 If $20 < \omega < 50$

then, $$G(s)H(s) = \frac{5 \times \frac{s}{20} \times 1}{\frac{s}{10} \times 1} = 2.5$$

$$M = 20 \log_{10} 2.5$$

Case: 4 If $50 < \omega < 100$

then, $$G(s)H(s) = \frac{5 \times \frac{s}{20} \times \frac{s}{50}}{\frac{s}{10} \times 1}$$

$$G(s)H(s) = \frac{5s}{100}$$

$$G(j\omega)H(j\omega) = \frac{5j\omega}{100}$$

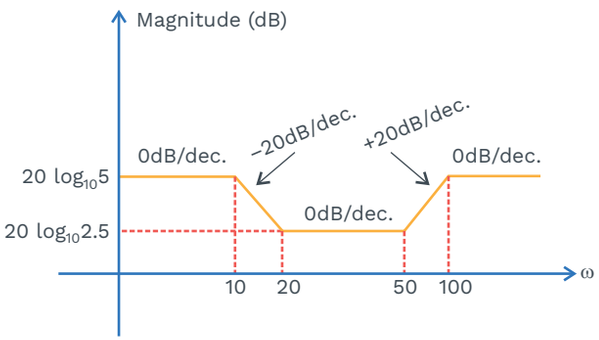
$$M = 20 \log 0.05 + 20 \log_{10}\omega$$

Case: 5 - If $\omega > 100$

then, $$G(s)H(s) = \frac{5 \times \frac{s}{20} \times \frac{s}{50}}{\frac{s}{10} \times \frac{s}{100}} = 5$$

$$M = 20 \log_{10} 5$$

Asymptotic bode diagram is shown below-




- At every pole, the slope changes by –20dB/decade.
- At every zero, the slope changes by +20dB/decade.

**6. Standard Second Order System-**

Consider the second order system having transfer function as -

$$T(s) = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

Put $s = j\omega$,

$$T(j\omega) = \frac{\omega_n^2}{-\omega^2 + 2j\xi\omega_n\omega + \omega_n^2}$$

$$T(j\omega) = \frac{\omega_n^2}{\left(\omega_n^2 - \omega^2\right) + 2j\xi\omega_n\omega}$$

$$T(j\omega) = \frac{1}{\left[1 - \left(\frac{\omega}{\omega_n}\right)^2\right] + 2j\xi\frac{\omega}{\omega_n}}$$

Magnitude, $\left|T\left(j\omega\right)\right| = \frac{1}{\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + 4\xi^2\frac{\omega^2}{\omega_n^2}}}$

Magnitude in dB -

$$M = 20\log\left|T(j\omega)\right| = -20\log_{10}\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + \left(2\xi\frac{\omega}{\omega_n}\right)^2}$$

**Case: 1 At very low frequencies**

$\omega << \omega_n$

$\frac{\omega}{\omega_n} << 1$

$\Rightarrow \quad M\left(dB\right) = -20\log_{10}\sqrt{1} = 0\,dB$

**Case: 2 Mid frequency region (at corner frequency)**

$\omega = \omega_n$

$$M = \frac{1}{\left[1 + \left(\frac{\omega}{\omega_n}\right)^4 - 2\left(\frac{\omega}{\omega_n}\right)^2 + 4\xi^2\left(\frac{\omega}{\omega_n}\right)^2\right]^{1/2}}$$

$$M = \frac{1}{2\xi}$$

Errors at corner frequency

$$= \pm 20 \log \sqrt{\left[1-\left(\frac{\omega_n}{\omega_n}\right)^2\right]^2 + \left(2\xi\frac{\omega_n}{\omega_n}\right)^2}$$

$$= \pm 20 \log 2\xi$$

**Case: 3 At very high frequencies**

$$\omega >> \omega_n$$

$$\frac{\omega}{\omega_n} >> 1$$

$$M = -20 \log_{10} \sqrt{\left(\frac{\omega^2}{\omega_n{}^2}\right)^2} = -20 \log_{10} \left(\frac{\omega}{\omega_n}\right)^2$$

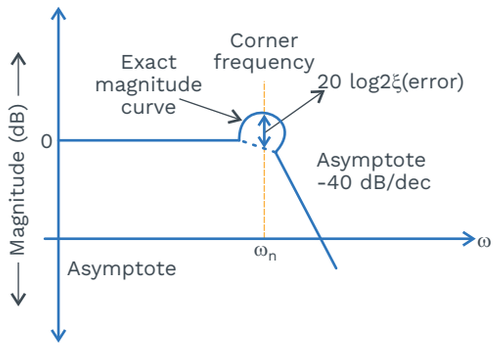
$$M = -40 \log_{10} \frac{\omega}{\omega_n}$$

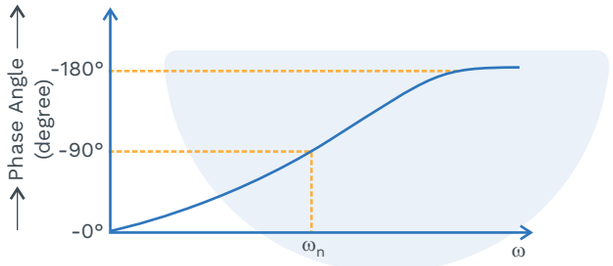
So, it is a straight line having a slope of –40dB/decade.

**Phase Angle Plot-**

$$\phi = \angle T(j\omega) = -\tan^{-1}\left[\frac{2\xi\frac{\omega}{\omega_n}}{1-\left(\frac{\omega}{\omega_n}\right)^2}\right]$$

1. For low frequency region –
   $\omega << \omega_n$
   $\phi = 0°$
2. At mid frequency region -
   $\omega = \omega_n$
   $\phi = -\tan^{-1}(\infty) = -90°$
3. At high frequency region -
   $\omega >> \omega_n$
   $\phi = -180°$




**Initial slope of bode plot -**

Let us consider, $G(s)\,H(s) = \frac{K}{s^N}$

Put $s = j\omega$,

$$G(j\omega)\,H(j\omega) = \frac{K}{(j\omega)^N}$$

In decibel, the magnitude is –

$$20\log_{10}\left|G(j\omega)\,H(j\omega)\right| = 20\log_{10}\left|\frac{K}{(j\omega)^N}\right|$$

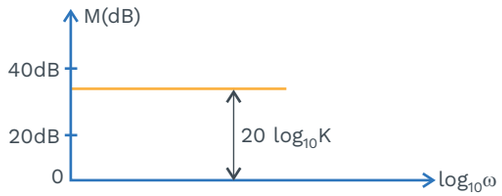
$$= 20\log_{10}K - 20N\log_{10}\omega \qquad \text{.... (I)}$$

Where, N denotes the type of the transfer function.
Therefore, the slope of bode plot is decided by the type of a system.

**1. Type-0 system (N = 0) -**

$$20\log_{10}\left|G(j\omega)\,H(j\omega)\right| = 20\log_{10}K$$

This is a straight line which can be plotted as-




**2. Type-1 system (N = 1) -**

For type 1 system, put N = 1 in equation (I) -

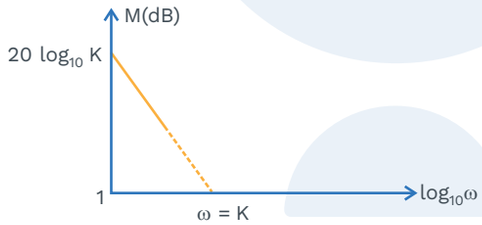
$$20 \log_{10} |G(j\omega)\, H(j\omega)| = 20 \log_{10} K - 20 \log_{10} \omega$$

The above equation is a straight line with slope of –20 dB/decade.
Intersection with 0dB axis is-

$$0 = 20 \log_{10} K - 20 \log_{10} \omega$$

$$K = \omega$$

Thus, the graph intersects 0dB axis at $\omega = K$, the graph can be drawn as –




**3. Type-2 System (N = 2)-**

Put N = 2 in equation (I) –

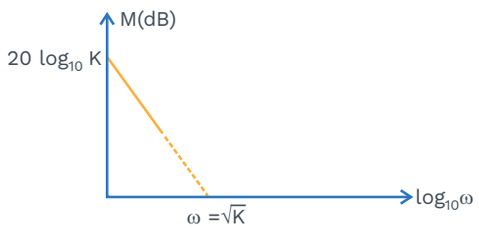
$$20 \log_{10} |G(j\omega)H(j\omega)| = 20 \log_{10} K - 40 \log_{10} \omega$$

The graph for the above equation has a straight line with slope of –40dB/decade and intersection with 0dB axis is given by –

$$0 = 20 \log_{10} K - 40 \log_{10} \omega$$

$$20 \log_{10} K = 20 \log_{10} \omega^2$$

Thus, graph intersects 0 dB axis at $\omega = \sqrt{K}$, which can be drawn as –



**Static error coefficients from initial slope of Bode plot -**

In sinusoidal form, static error coefficients can be described as –

**(a) Static position error constant ($K_p$) -**

Static position error is defined for type zero system and is given by –

$$K_p = \lim_{\omega \to 0} G(j\omega)\, H(j\omega)$$

Magnitude of $G(j\omega)H(j\omega)$ of type zero system is 20 log K which is $K_p$.

$$\therefore \quad K_p = 20 \log K$$

**Gray Matter Alert!!!**

- Bode plot is called as an asymptotic approximation.
- Asymptote is a line which touches the curve at infinite.

**(b) Velocity error coefficient ($K_v$)-**

Velocity error coefficient is defined for type one system.

$$K_V = \lim_{\omega \to 0} (j\omega)\, G(j\omega)\, H(j\omega)$$

So, static velocity error constant is-

$$K_v = \omega$$

**(c) Acceleration error constant ($K_a$) -**

$K_a$ is defined for type two system and can be written as –

$$K_a = \lim_{\omega \to 0} (j\omega)^2\, G(j\omega) H(j\omega)$$

For type-2 system, initial slope of bode plot is –40dB/decade.

$$20 \log_{10} \left| G(j\omega)\, H(j\omega) \right| = 20 \log_{10} K - 40 \log_{10} \omega$$

Intersection with 0dB line,

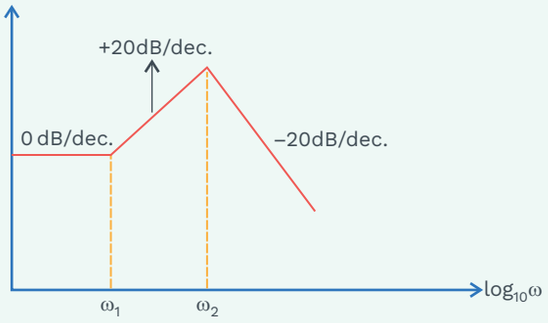
$$0 = 20 \log_{10} K_a - 40 \log_{10} \omega$$

$$\omega = \sqrt{K_a}$$

$$K_a = \omega^2$$

**Previous Year's Questions**

The asymptotic bode plot of transfer function is given below. In the frequency range shown, the transfer function has

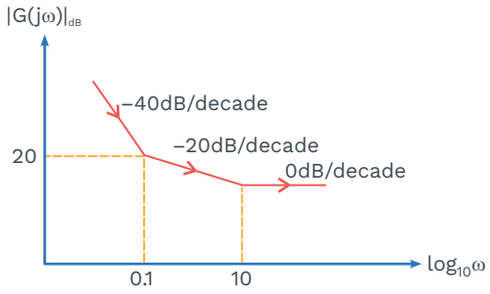



(a) 3 poles and 1 zero
(b) 1 pole and 2 zero
(c) 2 poles and 1 zero
(d) 2 poles and 2 zero

**Answer:** (c)

**Example - 16**

Find the transfer function for the system whose bode plot is shown-




**Solution:**

From the given graph -

Initial slope is –40 dB/decade which means type of system is 2.

There are two zeros at 0.1 rad/sec and 10 rad/sec.

System transfer function -

$$G(s)H(s) = \frac{K\left(1+\frac{s}{0.1}\right)\left(1+\frac{s}{10}\right)}{s^2}$$

At $\omega$ = 0.1 rad/sec,
Magnitude (dB) = 20dB

$$G(s)H(s) = \frac{K \times 1 \times 1}{(0.1)^2} = 100\ K$$

Magnitude in decibel -

$$20 \log 100\ K = 20$$

$$100\ K = 10$$

$$K = 0.1$$

Hence, $$G(s)H(s) = \frac{0.1\left(1+\frac{s}{0.1}\right)\left(1+\frac{s}{10}\right)}{s^2}$$

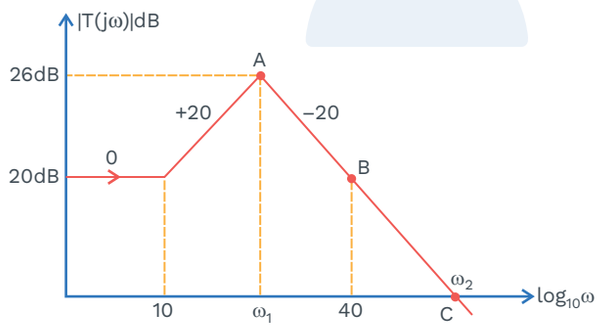
**Previous Year's Questions**

For the transfer function -

$$G(s) = \frac{5(s+4)}{s(s+0.25)(s^2+4s+25)}$$

The values of the constant gain term and the highest corner frequency of the bode plot respectively are -
(a) 3.2 and 5 (b) 3.2 and 4.0
(c) 16.0 and 4.0 (d) 16.0 and 5.0
**Answer:** (a)

**Example - 17**
For the asymptotic bode magnitude plot shown below, determine the value of $\omega_1$ and $\omega_2$.




**Solution:**
From the given plot -
Slope between 10 and $\omega_1$ is +20dB/decade.

$$\therefore \quad 20 = \frac{26-20}{\log\ \omega_1 - \log 10}$$

$$\frac{6}{20} = \log \frac{\omega_1}{10}$$

$\omega_1$ = 20 rad/sec.

and slope between $\omega_1$ and $\omega_2$ is –20dB/decade.

$$-20 = \frac{0 - 26}{\log \omega_2 - \log \omega_1}$$

$$\log \frac{\omega_2}{\omega_1} = 1.3$$

$$\omega_2 = 400 \text{ rad/sec}$$

**Example - 18**

Draw the bode plot for the open loop transfer function for the unity feedback system is $G(s) = \frac{50}{(s+1)(s+2)}$. Also determine the stability using gain margin and phase margin parameter.

**Solution:**

Given,

$$G(s) = \frac{50}{(s+1)(s+2)}$$

Transfer function in time constant form is-

$$G(s) = \frac{50}{2(1+s)\left(1+\frac{s}{2}\right)}$$

$$G(s) = \frac{25}{(1+s)\left(1+\frac{s}{2}\right)}$$

Put s = jω,

$$G(j\omega) = \frac{25}{(1+j\omega)(1+0.5j\omega)}$$

The type of system is 0, so the initial slope of the bode plot is at 0dB/decade and the initial ordinate is -

$$20 \log_{10} 25 = 27.9 \text{ dB}$$

Then the two corner frequencies are –

$$\omega_1 = 1 \text{ rad/sec}$$

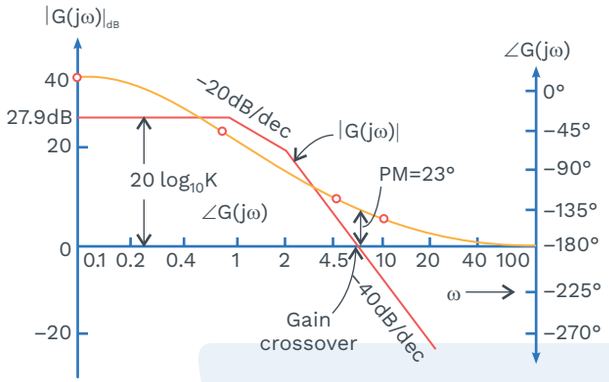
$$\omega_2 = \frac{1}{0.5} = 2 \text{ rad/sec}$$

Slope of magnitude plot after $\omega_1$ = (0 – 20)

$\omega_1$ = –20 dB/decade

And slop of magnitude plot after $\omega_2$ = (–20–20) = –40 dB/decade

The Bode plot of $|G(j\omega)|$ and $\angle G(j\omega)$ versus $\omega$ (log scale) is-




Phase angle is given by -

$$\angle G(j\omega) = -\tan^{-1}\omega - \tan^{-1}(0.5\,\omega)$$

For frequency range $\omega$ = 0.1 rad/sec and 10 rad/sec is calculated as –

| $\omega$(rad/sec) | 0.1 | 1 | 5 | 10 |
|---|---|---|---|---|
| $\angle G(j\omega)$ | –8.5 | -71.5 | -146.8 | -163 |

At the gain crossover frequency, the phase angle is –157°.
Thus, phase margin is,

$$PM = 180° + (-157°) = 23°$$

- Phase curve is asymptotic to –180° line at high frequencies and gain at high frequencies is negative. So, gain margin is infinite (positive).
- Since, gain margin and phase margin both are positive. Hence, the system is stable.

**Example - 19**

The open loop transfer function of a system is $\frac{10(s-1)^5 e^{-3s}}{s(s+2)(s-4)}$. Find the high frequency slope of the bode magnitude plot.

**Solution:**

Given,

$$G(s)\,H(s) = \frac{10(s-1)^5 e^{-3s}}{s(s+2)(s-4)}$$

Put s = jω,

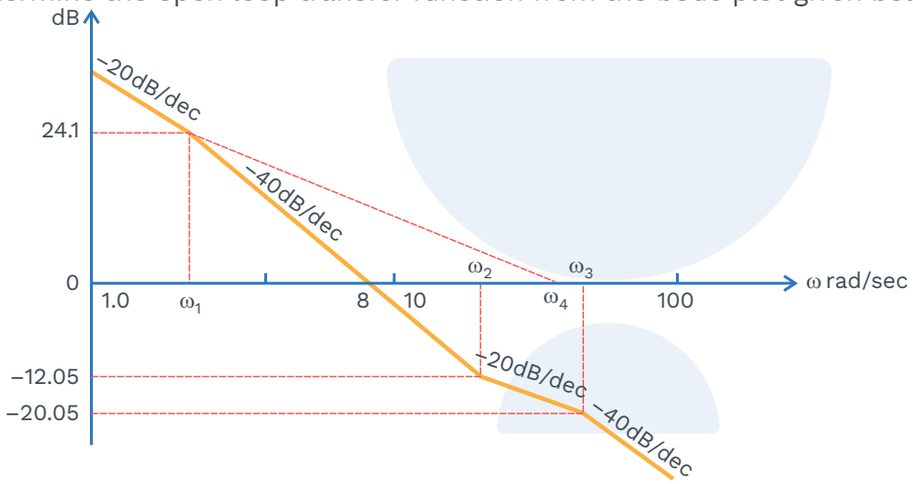
$$G(j\omega)\,H(j\omega) = \frac{10(j\omega - 1)^5\, e^{-3j\omega}}{j\omega(j\omega + 2)(j\omega - 4)}$$

Every zero will contribute +20dB/decade and each pole will contribute –20dB/decade slope and there is five zeros and three poles. So, the slope at high frequency is–

Slope = 5 × 20 – 3 × 20 = 40 dB/decade.

**Example - 20**

Determine the open loop transfer function from the bode plot given below.




**Solution:**

From the given plot -

Initial slope of the plot is –20dB/decade. So, the type of the system is 1

Intersection with 0dB line is at $\omega_4$-

$\therefore \quad K = \omega_4$

At $\omega_1$, slope of plot changes by –20dB/decade, the corresponding term of transfer function is $\frac{1}{1+sT_1}$

Where, $T_1 = \frac{1}{\omega_1}$

And the slope between $\omega_1$ and 8 rad/sec is –40dB/decade. which can be written as -

$$-40 = \frac{0 - 24.1}{\log 8 - \log \omega_1}$$

$\omega_1$ = 2 rad/sec

$$T_1 = \frac{1}{\omega_1} = \frac{1}{2} = 0.5 \text{ seconds}$$

At $\omega_2$, slope of plot changes by +20dB/decade. Hence, the corresponding term in the transfer function is $1 + sT_2$.

Where, $T_2 = \frac{1}{\omega_2}$

The slope between $\omega_1$ and $\omega_2$ is –40dB/decade and is given by -

$$-40 = \frac{12.05 - 24.1}{\log \omega_2 - \log \omega_1}$$

$$\frac{\omega_2}{\omega_1} = 2 \Rightarrow \omega_2 = 4 \text{ rad / sec}$$

$$T_2 = \frac{1}{\omega_2} = \frac{1}{4} = 0.25 \text{ seconds}$$

At $\omega_3$, the slope of plot changes by –20 dB/decade. Hence, the corresponding term of transfer function is $\frac{1}{1 + sT_3}$.

Where, $T_3 = \frac{1}{\omega_3}$

The slope between $\omega_2$ and $\omega_3$ is –20dB/decade.

$$-20 = \frac{-20.05 - (-12.05)}{\log \omega_3 - \log \omega_2}$$

and $T_3 = \frac{1}{\omega_3} = \frac{1}{10} = 0.1 \text{ seconds}$

Slope between $\omega_4$ and $\omega_1$ is –20dB/decade.

$$-20 = \frac{0 - 24.1}{\log \omega_4 - \log \omega_1}$$

$\omega_4 = 16\ \omega_1 = 16 \times 2 = 32$ rad/sec

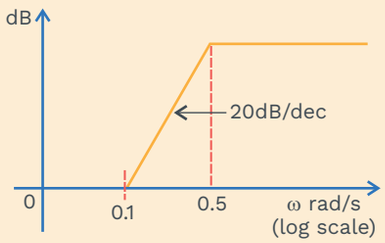
and $K = \omega_4 = 32$

Thus, the overall transfer function is –

$$T(s) = \frac{32(1 + 0.25s)}{s(1 + 0.5s)(1 + 0.1s)}$$

## PRACTICE QUESTIONS

**Q1** **What will be the frequency (in rad/sec) at maximum phase angle of the phase lead network whose asymptotic bode magnitude plot with its pole and zero on the left half of s-plane is shown in the mentioned figure?**




(a) $\frac{3}{\sqrt{10}}$ (b) $\frac{1}{\sqrt{20}}$

(c) $\frac{1}{20}$ (d) $\frac{1}{30}$

**Sol** (d)

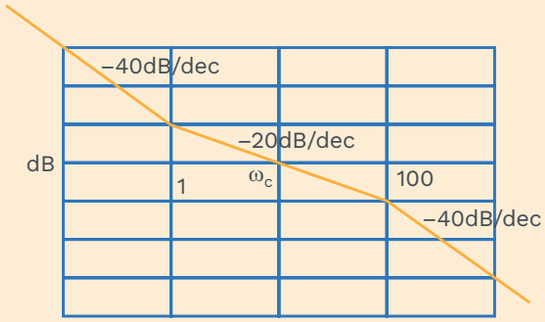
**Q2** **The open loop transfer function of a negative feedback system is $G(s)\,H(s) = \frac{2(s+1)}{s^2}$. The phase margin of the system (in degrees, rounded off to one decimal place) is_____.**

**Sol** (65.5)

**Q3** **In a control system with unity gain feedback, the transfer function of the open loop-gain function is $L(s) = \frac{9e^{-0.1s}}{s}$. The phase margin of the loop-gain function L(s) is _____ degrees.**

**Sol** (38.43)

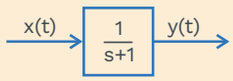
**Q4** **For an open loop control system, the asymptotic magnitude bode plot with K > 0 and all the poles and the zeros on the left-hand side of the s-plane is shown below. The plot is completely symmetric about $\omega_c$ then the minimum absolute phase angle contribution by G(s) is-**




**(a) 78.6°** **(b) 90°** **(c) 101.4°** **(d) 180°**

**Sol** **(d)**

**Q5** **If the input signal for the system given below is x (t) = (sin t). Then in steady-state the response of the system is-**

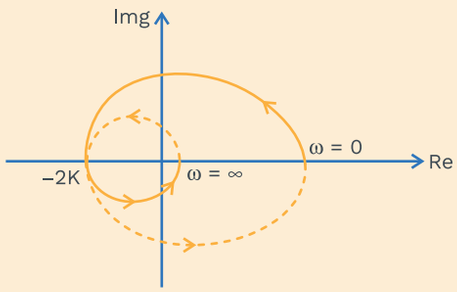



**(a)** $\frac{1}{\sqrt{2}}\sin\left(t-\frac{\pi}{4}\right)$ **(b)** $\frac{1}{\sqrt{2}}\sin\left(t+\frac{\pi}{4}\right)$

**(c)** $\frac{1}{\sqrt{2}}e^{-t}\sin t$ **(d) sin t – cos t**

**Sol** **(a)**

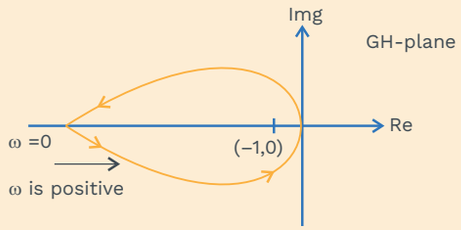
**Q6** **The Nyquist plot of a control system is shown in the figure below. It is known that the number of open loop poles having positive real part is two. For the given system to be stable, the range of positive values of K will be**




**(a) K > 5** **(b) K < 2** **(c) K > 0.5** **(d) K < 0.5**

**Sol** **(c)**

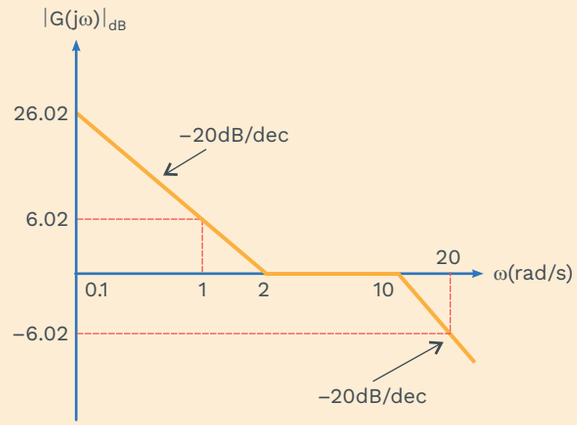
**Q7** **At what stage will be the closed loop system, when the G(s)H(s) has the pole on right hand side for the mentioned Nyquist plot of the open loop transfer function G(s)H(s) of a system?**




**(a) The control system is always stable**
**(b) The control system is unstable with one closed loop pole on right hand side**
**(c) The control system is unstable with two closed loop poles on right hand side**
**(d) The control system is unstable with three closed loop poles on right hand side**

**Sol** **(a)**

**Q8** **For a minimum phase a system, the asymptotic magnitude plot is given below.**




**If the system is connected in a unity negative feedback configuration, then the steady state error of the closed loop system to a unit ramp input is_____**

**Sol** **(0.5)**

**Q9** **Sketch the bode plot of a closed loop system which has the open loop transfer function**

$$G(s)\,H(s) = \frac{2e^{-sT}}{s(1+s)(1+0.5s)}$$

**Determine the maximum value of T for the system to be stable.**

**Sol** **(0.168 sec)**

**Q10** **A unity feedback control system has an open loop transfer function as-**

$$G(s) = \frac{K}{s(1+0.5s)(1+0.25s)}$$

**If K = 1, determine the value of K if gain margin is 8dB.**

**Sol** **K = 2.23**